

# BIM PORTFOLIO





# KATRAVATHU RAHUL NAIK

 **+916300075601**

 **katravathurahul@gmail.com**

 **<https://www.linkedin.com/in/k-rahul-naik/>**

 **<https://github.com/Katravathu9>**

## technical skills



# **BIM TECHNICAL SKILLS**

- Planning, Designing & Detailing of Residential, Commercial & Educational structures, Preparation of 3d Models.
- BIM Modelling for specific design disciplines.
- Hands-on experience in using BIM software's such as Revit, and AutoCAD.
- Ensure Construction document accuracy based on Architecture BIM Modelling.
- Collaborate and Coordinate with disciplines design-changes and model challenges.
- Acquired knowledge of BIM Execution Plan (BEP), COBie and BIM project workflow criteria.
- Understanding concepts like LOD, and Dimensions of BIM, Parameters, CAD to BIM, Scan to BIM & BIM 360.
- Acquired knowledge of Work Sharing, and how to create Central & Local files.
- Creation of Revit families (Parametric & Non-Parametric).
- Realistic Renderings and walkthroughs of Architectural Models on Escape.

# **SOFT SKILLS**

- Good Planning, designing & Visualization Skills.
- Keen to learn new technology.
- Quick learner, Curious to learn new things.
- Problem-solving ability.
- Hardworking, Honest, Self motivated, Discipline.
- Good Verbal Communication skills (English, Hindi, Telugu).
- Ability to cope with failures and try to learn from them

# **Projects**

1)Gated Community

2)1BHK Design

3)2BHK Home Design

4)Topaz Hostel

5) G+2 RESIDENTIAL BUILDING

6) CAPITAL Height

## 1)GATED COMMUNITY

- Designed a G+6 gated community project using 2D AutoCAD drafting and 3D Revit BIM modeling, including foundation layout and structural components, reducing design errors by 25% through coordinated modeling.
- Prepared cost estimation and BOQ with quantity takeoff from Revit models, ensuring 100% compliance with RERA regulations, Vastu principles, and local building by-laws, improving approval efficiency by 20%.
- Executed accurate structural detailing and submission drawings, minimizing rework by 30% and ensuring timely project submission as per statutory authority requirements.

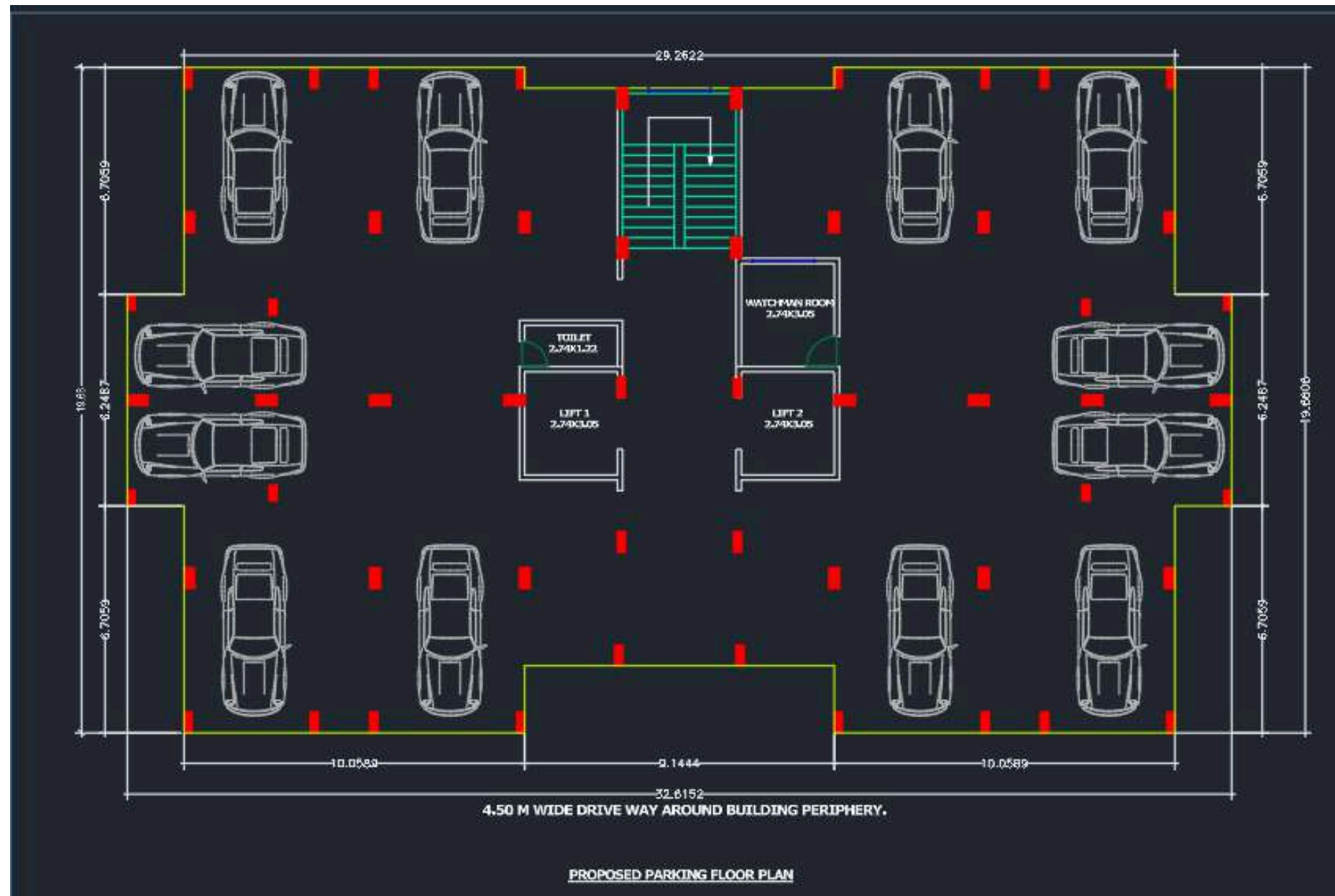






3D View





Parking Area and  
Main Entrance



Top view

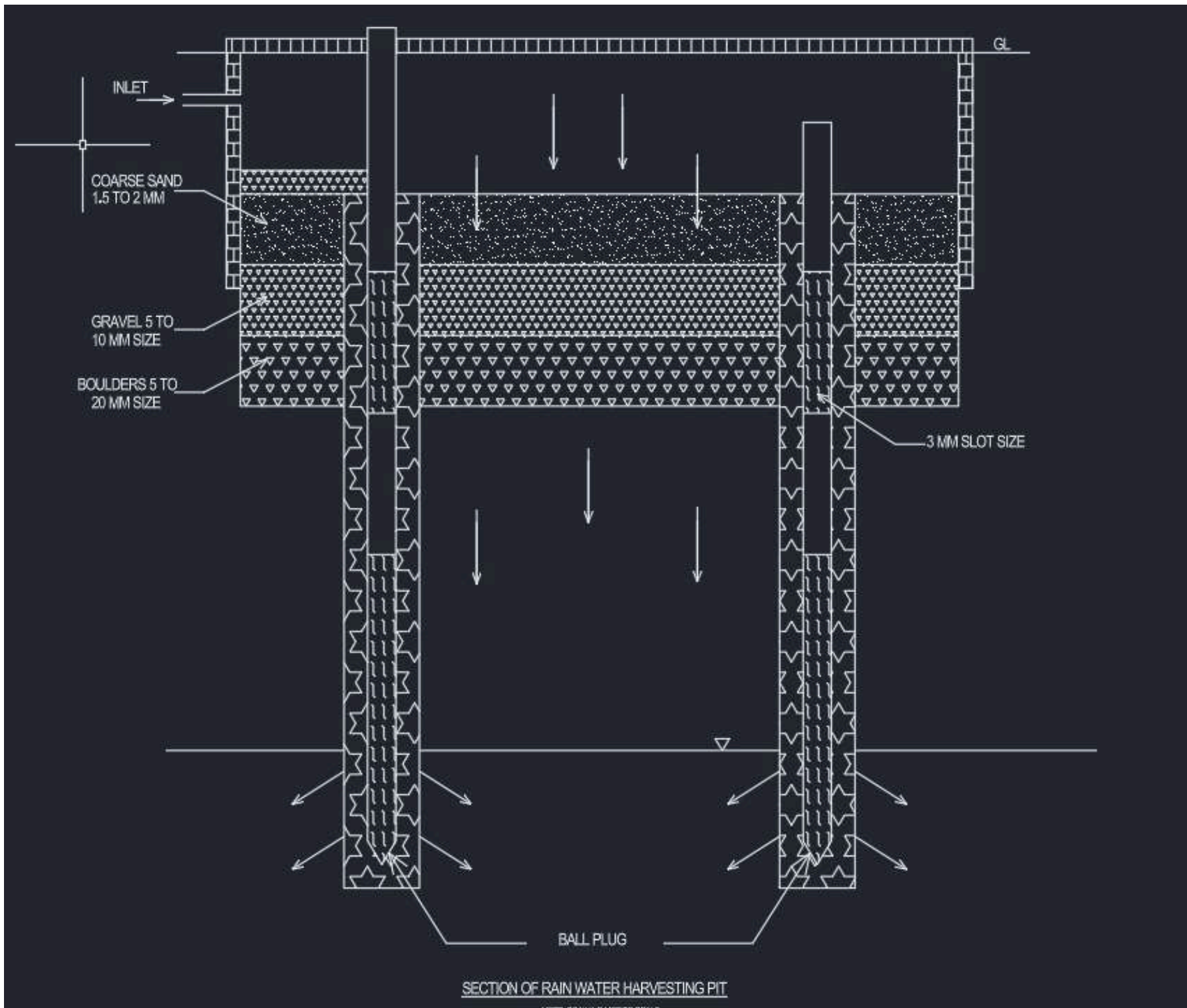




Front View

- This is a G+6 residential apartment building with a symmetrical elevation, central staircase ( $\approx 1.5\text{--}2.0$  m wide), balconies projecting about 1.2 m, typical floor height  $\approx 3.0$  m, and total building height  $\approx 18\text{--}20$  m including parapet.
- The section shows RCC framed structure with columns ( $\approx 0.3$  m  $\times$  0.45 m), beams ( $\approx 0.23$  m  $\times$  0.45 m), slab thickness  $\approx 0.125\text{--}0.15$  m, parking at ground floor (clear height  $\approx 2.4\text{--}3.0$  m), and overall width  $\approx 15\text{--}18$  m.





#### REFERENCE

**PROPOSED SHOWN AS**



**RAIN WATER HARVESTING PITS**  
(OF 6.80 X 2.00 X 1.50 M) SHOWN AS  
(2 No.s WITH A VOLUME OF 10.20 Cum EACH; TOTAL = 20.40 Cum.)

R.W.H.P

**MORTGAGE AREA SHOWN AS**



**DRIVE WAY SHOWN AS ( IN STILT FLOOR)**



#### **MORTGAGE AREA INFORMATION:**

\* THE BUILT-UP AREA OF 166.45 Sq. Mts. (10.07 %) AT SECOND FLOOR REQUIRED TO BE HAND OVER TO THE SANCTIONING AUTHORITY BY WAY OF A NOTARISED AFFIDAVIT.

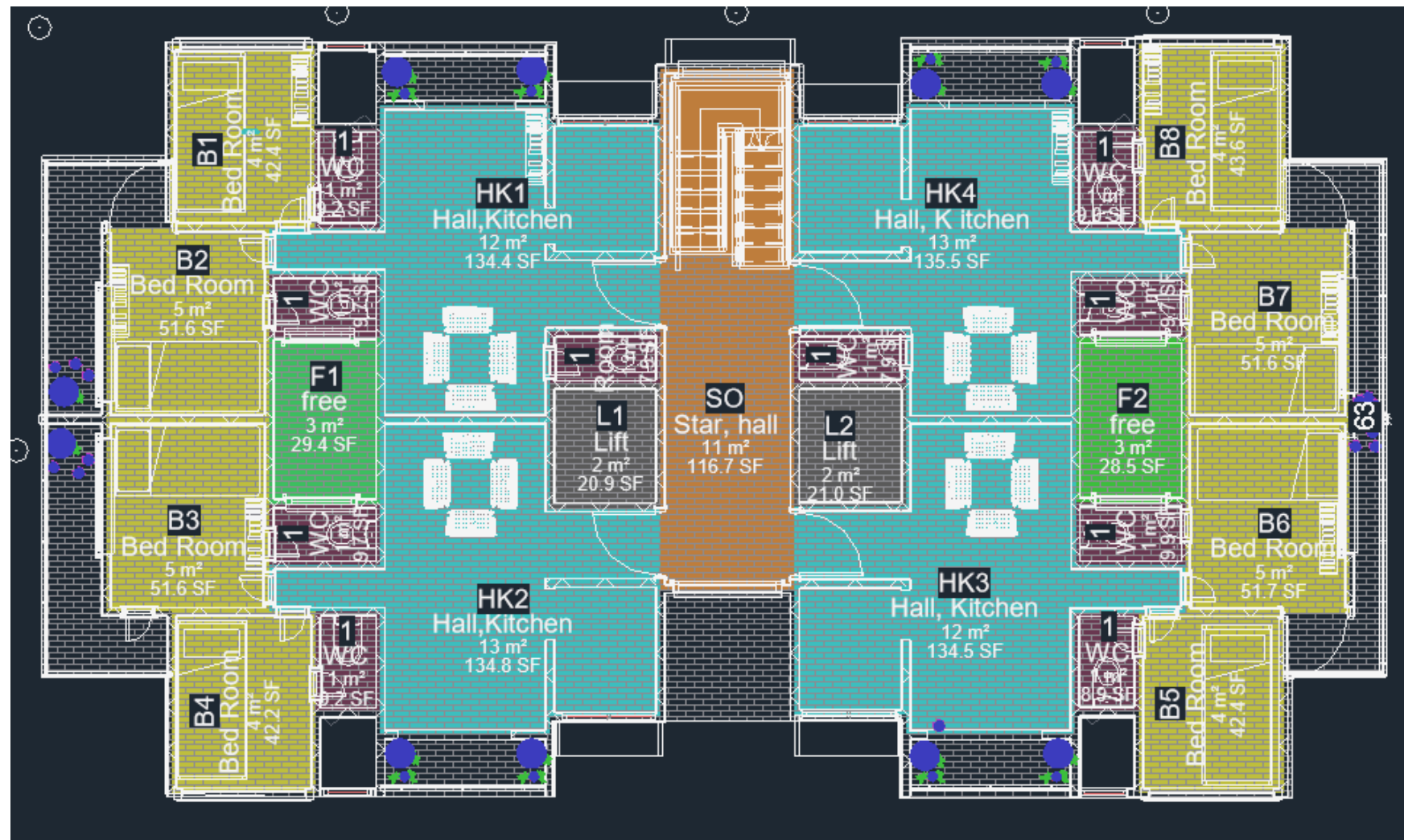
**MORTGAGE AREA SHOWN AS**



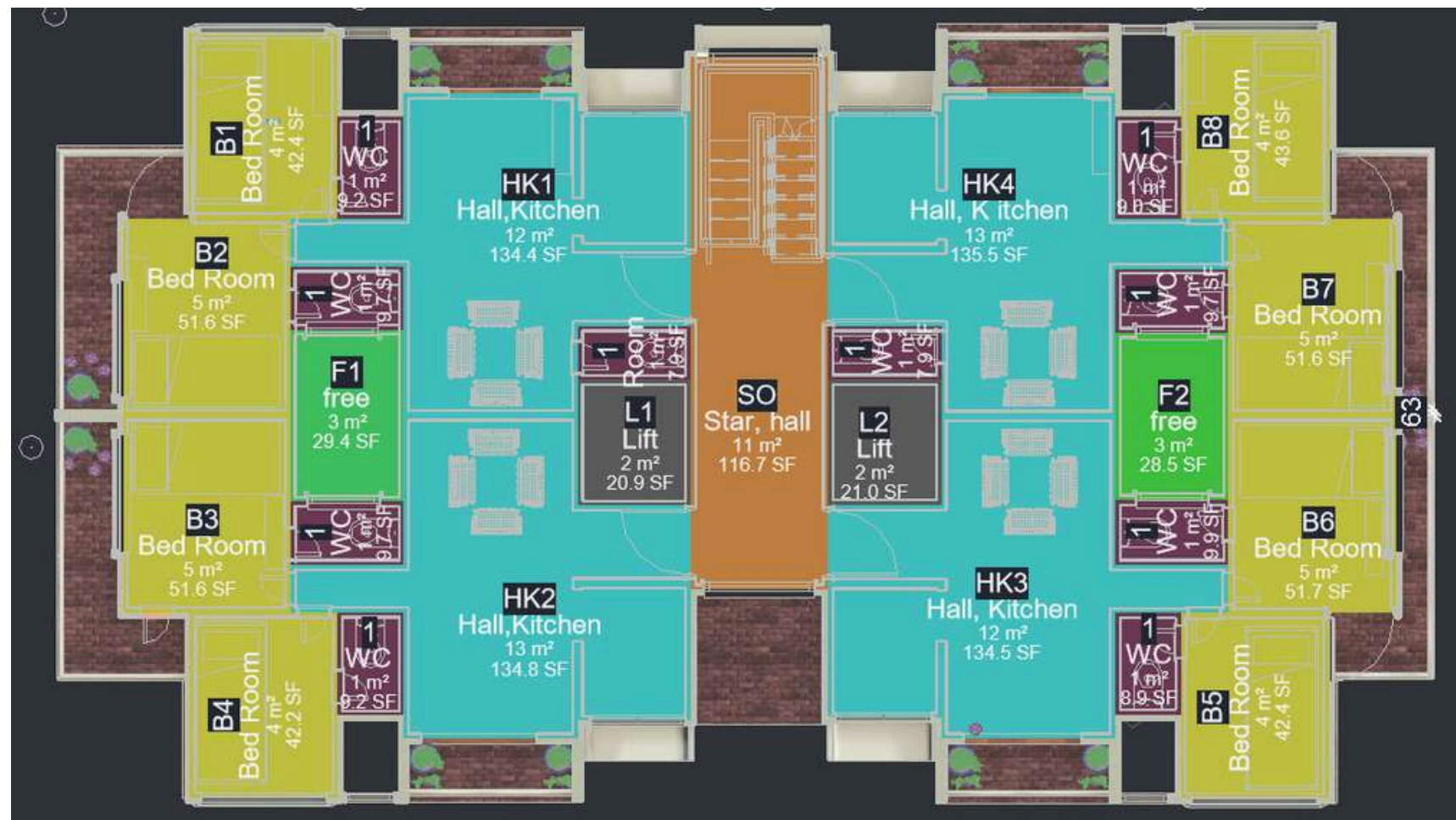
Section of Rain Water Harvesting Pit

**UNITES IN METTERS**





### Color Explanation in Plan (Path / Zoning):

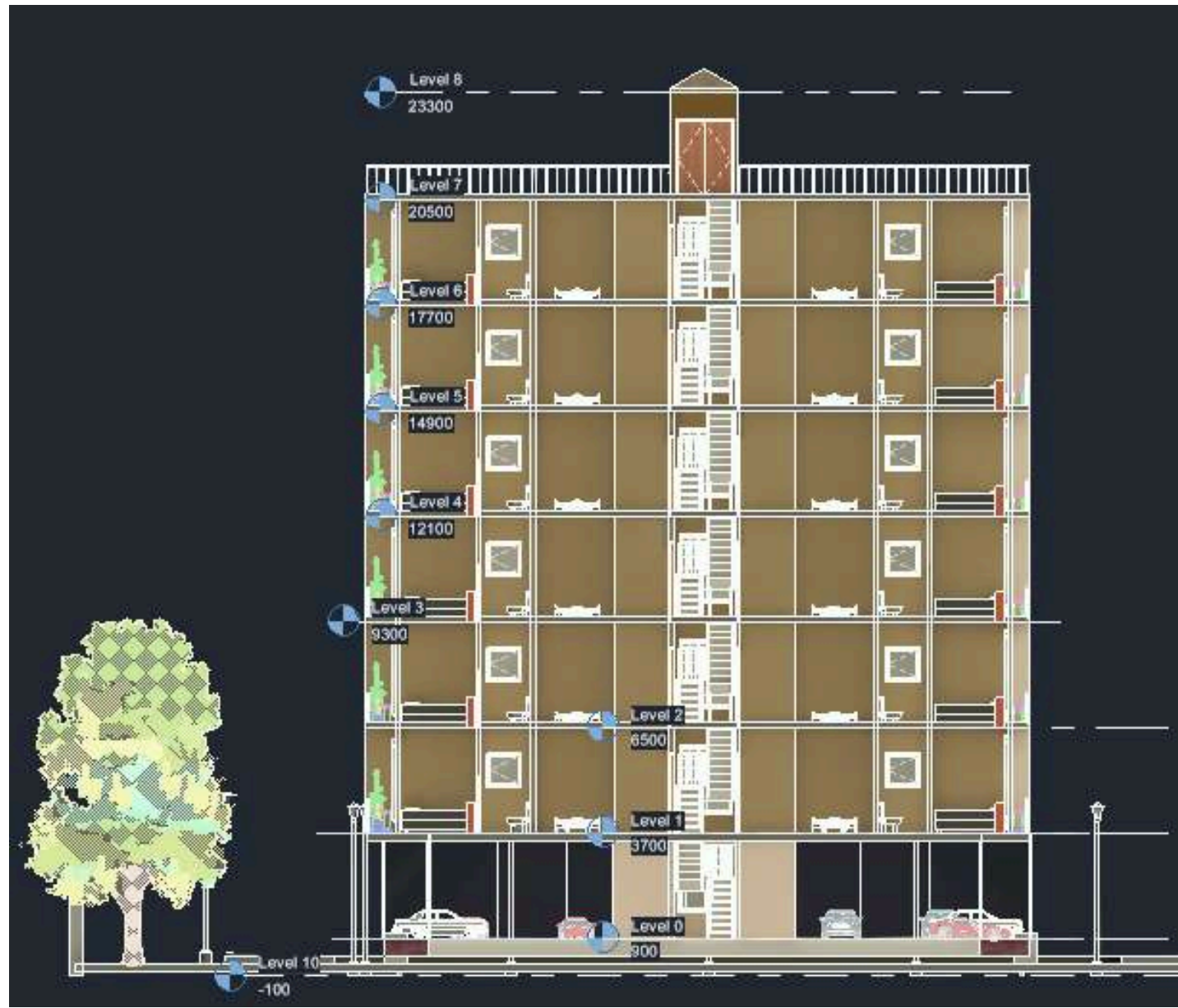


- Light Blue (HK1, HK2, HK3, HK4): Hall + Kitchen areas (main living spaces).
- Yellow (B1–B8): Bedrooms.
- Green (F1, F2): Free / Open utility spaces ( $\approx 3 \text{ m}^2$ ).
- Brown (SO): Staircase lobby / common circulation area.
- Dark Grey: Lift areas (L1, L2).
- Purple / Maroon: \*\* Toilets / W.C. areas.
- White Lines: Doors, windows, and internal partitions.









## 3D & Front View





In this 3D residential design, colours can be related to human comfort, emotions, and friendly connection with nature (animals & environment):

- Earthy Brown / Beige (Building Walls): Represents stability and warmth — like soil and natural habitat — creating a safe and comfortable feeling for humans and animals.
- Terracotta / Red Roof: Symbol of shelter and protection, similar to traditional homes that feel secure and welcoming.
- Green Areas (Landscape): Connects to nature, trees, birds, and small animals — promotes harmony, freshness, and healthy living for people.
- White / Light Railings & Frames: Shows cleanliness, peace, and openness — encouraging friendly and positive human interaction.
- Dark Grey Roads: Represents movement and connectivity — safe pathways for people while separating vehicle zones from green areas.

Overall, the colour combination creates a friendly, natural, and human-centered environment, encouraging close relationships between residents, nature, and surrounding living beings.



**SPECIFICATIONS:**

|                                                   |   |               |
|---------------------------------------------------|---|---------------|
| MD - MAIN DOOR                                    | - | 1.20 X 2.10 M |
| D - DOOR                                          | - | 0.90 X 2.10 M |
| D1 - FLUSH DOOR                                   | - | 0.75 X 2.10 M |
| W1 - WINDOW                                       | - | 1.20 X 0.9 M  |
| W2 - WINDOW                                       | - | 1.83 X 1.20 M |
| W3 - WINDOW                                       | - | 1.20 X 1.20 M |
| W4 - WINDOW                                       | - | 1.20 X 1.20 M |
| W5 - WINDOW                                       | - | 1.20 X 0.76 M |
| V - TRANSOM WINDOW                                | - | 0.60 X 0.60 M |
| ST - SEPTIC TANK                                  |   |               |
| B.W.- BORE WELL                                   |   |               |
| R. W.H.P. - WATER HARVESTING PITS OF 1.50 M DEPTH |   |               |
| PLASTERING WITH C.M (1:5)                         |   |               |
| BRICK MASONRY WITH C.M (1:6)                      |   |               |

**AREA STATEMENT:**

|                                                          |   |               |
|----------------------------------------------------------|---|---------------|
| TOTAL SITE AREA (AS PER DOCUMENT)                        | - | 2200.15 Sq. M |
| TOTAL SITE AREA (AS ON GROUND)                           | - | 2194.67 Sq. M |
| PROPOSED PARKING FLOOR AREA                              | - | 572.57 Sq. M  |
| PROPOSED WATCHMAN ROOM AND TOILETS AREA AT PARKING FLOOR | - | 11.63 Sq. M   |
| PROPOSED FIRST FLOOR AREA                                | - | 514.95 Sq. M  |
| PROPOSED SECOND FLOOR AREA                               | - | 514.95 Sq. M  |
| PROPOSED THIRD FLOOR AREA                                | - | 514.95 Sq. M  |
| PROPOSED FOURTH FLOOR AREA                               | - | 514.95 Sq. M  |
| PROPOSED FIFTH FLOOR AREA                                | - | 514.95 Sq. M  |
| PROPOSED SIXTH FLOOR AREA                                | - | 514.95 Sq. M  |
| PROPOSED LIFT AREA AT PARKING FLOOR                      | - | 24.50 Sq. M   |
| PROPOSED STAIRCASE AREA @ PARKING FLOOR                  | - | 18.40 Sq. M   |
| REMAINING SITE AREA                                      | - | 330.58 Sq. M  |



| COLOR INDEX                   |  |
|-------------------------------|--|
| PLOT BOUNDARY                 |  |
| ABUTTING ROAD                 |  |
| PROPOSED WORK (COVERAGE AREA) |  |
| EXISTING (To be retained)     |  |
| EXISTING (To be demolished)   |  |

Color Index:  
PARKING CALCULATION:

| Parking Type  | Prop No. | Prop Area |
|---------------|----------|-----------|
| Other Parking | 1        | 572.57    |
| Total Area    | 1        | 572.57    |

MARGIN DETAIL:

| Building / Wing Name | Road Name         | Front Margin | Rear Margin | Side1 Margin | Side2 Margin |
|----------------------|-------------------|--------------|-------------|--------------|--------------|
| A-1 (CHAMARTI)       | 18.00 M WIDE ROAD | 19.7         | 6.10        | 6.10         | 6.10         |

|                                                  |                                                |
|--------------------------------------------------|------------------------------------------------|
| AREA STATEMENT: TENALI (CRDA)                    | VERSION NO.: 1.101.44                          |
|                                                  | VERSION DATE: 24/01/2023                       |
| PROJECT DETAIL :                                 |                                                |
| BA No. : -                                       | Proposed Use : Residential                     |
| Invoke Department : AP DTCP ULBs                 | Proposed Activity : Residential Apartment Bldg |
| Region : SANGLI REGION                           | Plot Near by Religious Structure : NA          |
| District : SANGLI                                | Land Use Zone : NA                             |
| Application Type : General Proposal              | Land Sub Use Zone : NA                         |
| Project Type : Building Permission               | Abutting Road Width : -                        |
| Nature of Development : New                      | Plot No. : -                                   |
| Location : Municipality under CRDA               | Survey No. : -                                 |
| Sub Location : New Areas / Approved Layout Areas | North : -                                      |
| ZDP : NA                                         | South : -                                      |
| Village Name : SANGALWADI                        | East : -                                       |
| OC Type : Fully OC                               | West : -                                       |

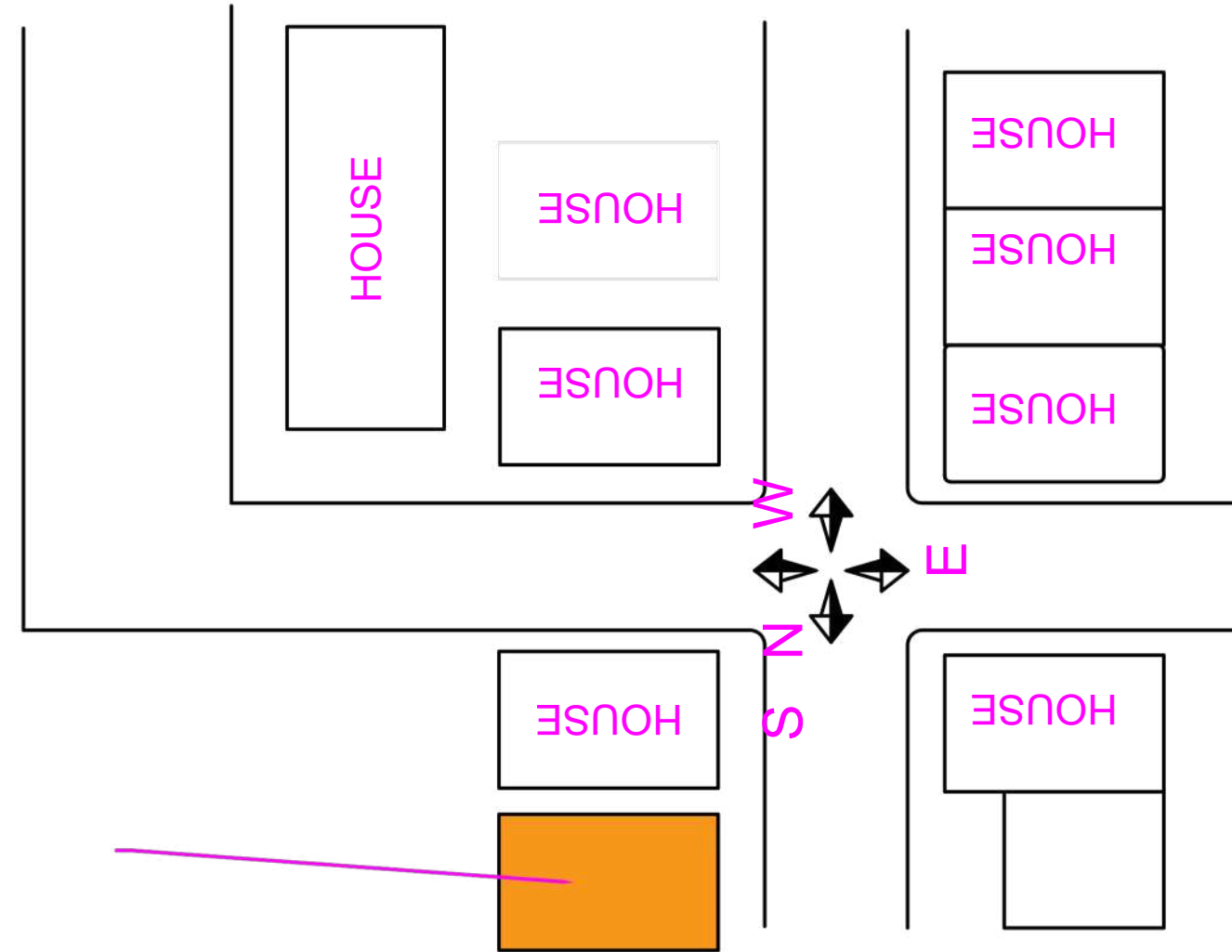
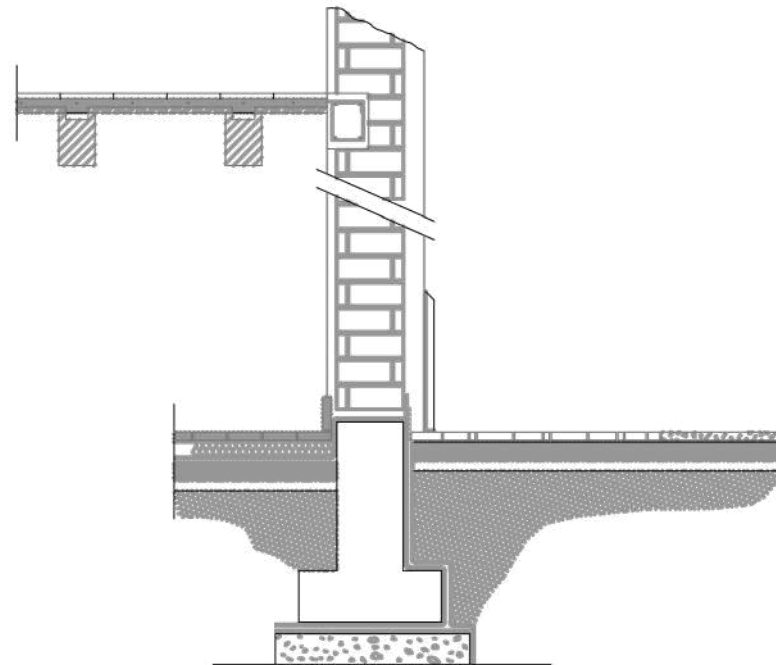
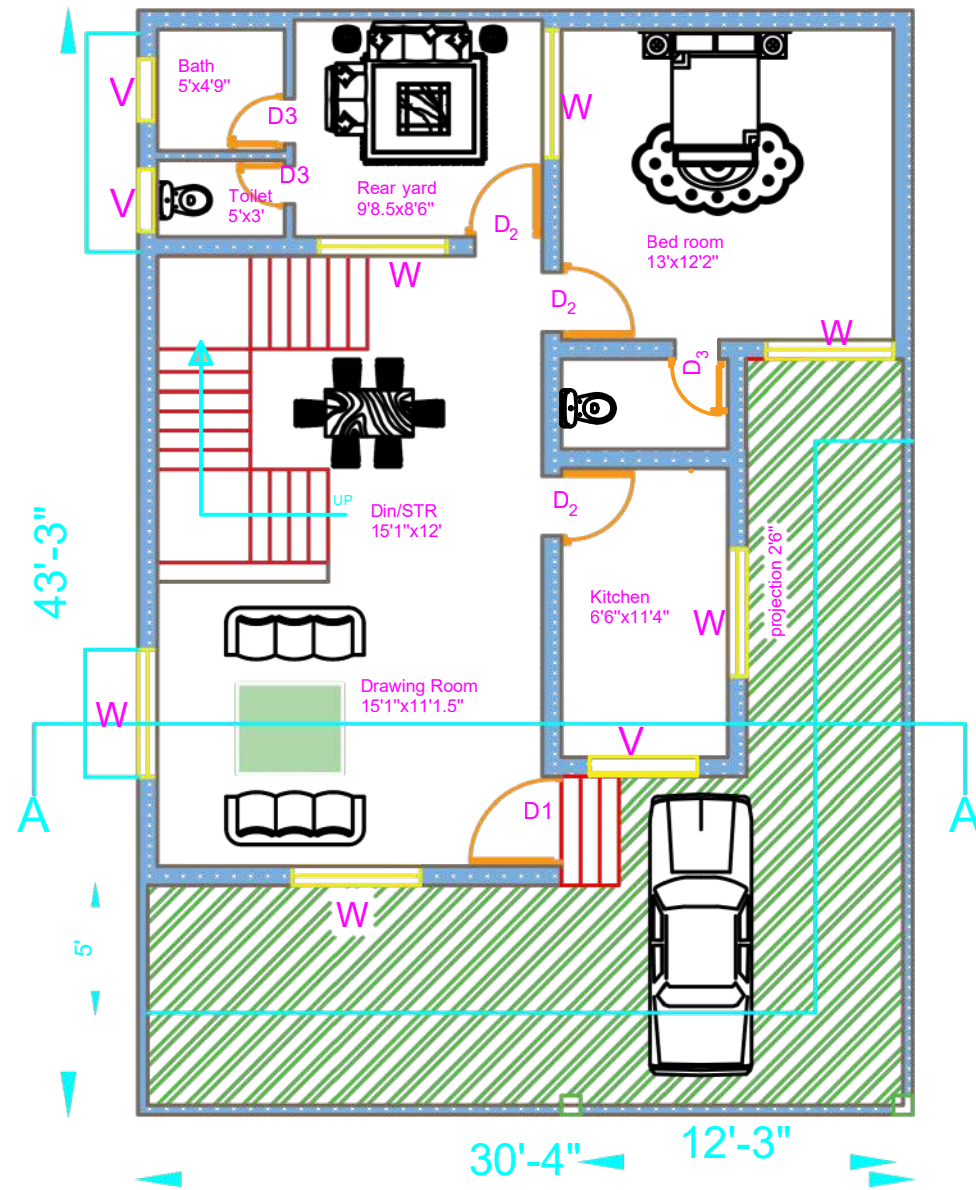
|                                         |                |         |
|-----------------------------------------|----------------|---------|
| AREA OF PLOT (Minimum)                  | (A)            | 2194.67 |
| NET AREA OF PLOT                        | (A-Deductions) | 2194.67 |
| AccessoryUse Area                       |                | 9.00    |
| Vacant Plot Area                        |                | 1622.1  |
| COVERAGE CHECK                          |                |         |
| Proposed Coverage Area ( 51.55 % )      |                | 572.57  |
| Total Prop. Coverage Area ( 51.55 % )   |                | 572.57  |
| Net BUA CHECK                           |                |         |
| Proposed OC Residential Net BUA         |                | 1652.90 |
| StairCase Area Add in Net BUA           |                | 20.76   |
| Lift Lobby Area Add in Net BUA          |                | 13.14   |
| Net BUA Area                            |                | 1682.40 |
| Total Net BUA Area                      |                | 1682.40 |
| BUILT UP AREA CHECK                     |                |         |
| Proposed BuiltUp Area                   |                | 2019.35 |
| SubStructure Area Added in BuiltUp Area |                | 9.00    |
| Total BuiltUp Area                      |                | 2028.35 |
| ARCH / ENGG / SUPERVISOR (Regd)         | OWNER          |         |
|                                         |                |         |
| DEVELOPMENT AUTHORITY                   | LOCAL BODY     |         |
|                                         |                |         |



## 2)1BHK Design

- Software Workflow: Use AutoCAD for 2D drafting and Revit for 3D BIM coordination ensuring all project units are set to Architectural (Feet/Inches) for precise on-site execution.
- RERA & Area Math: Utilize Excel to calculate the Net Carpet Area (inner wall-to-wall) as per RERA mandates, strictly excluding balconies and external wall thicknesses from the final saleable area.
- Legal Compliance: Design the unit to satisfy Building Bye-laws, maintaining minimum ceiling height of 9 feet and providing window ventilation equal to at least 10% of the floor area.
- Vastu Strategy: Position the Kitchen in the South-East and the Master Bedroom in the South-West, ensuring the main entrance is located in the North or East to align with traditional energy zones.





## Schedule of Door and Window

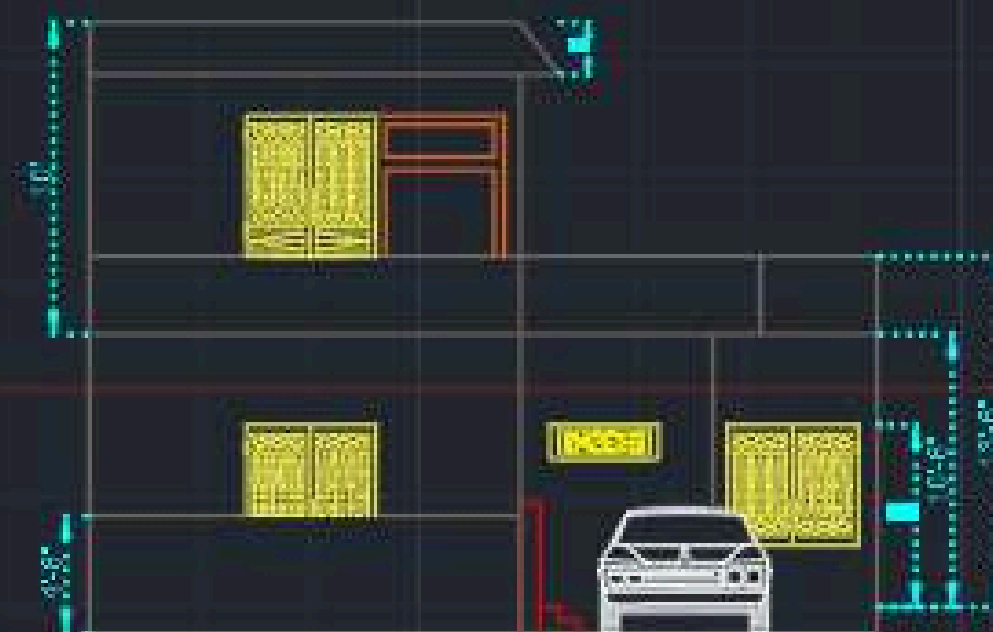
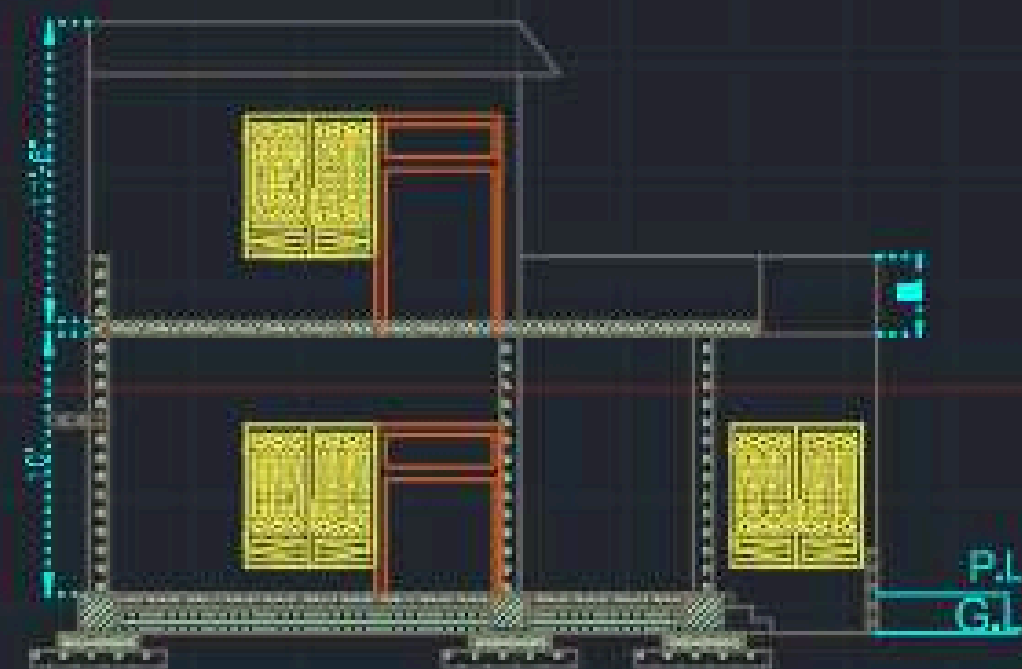
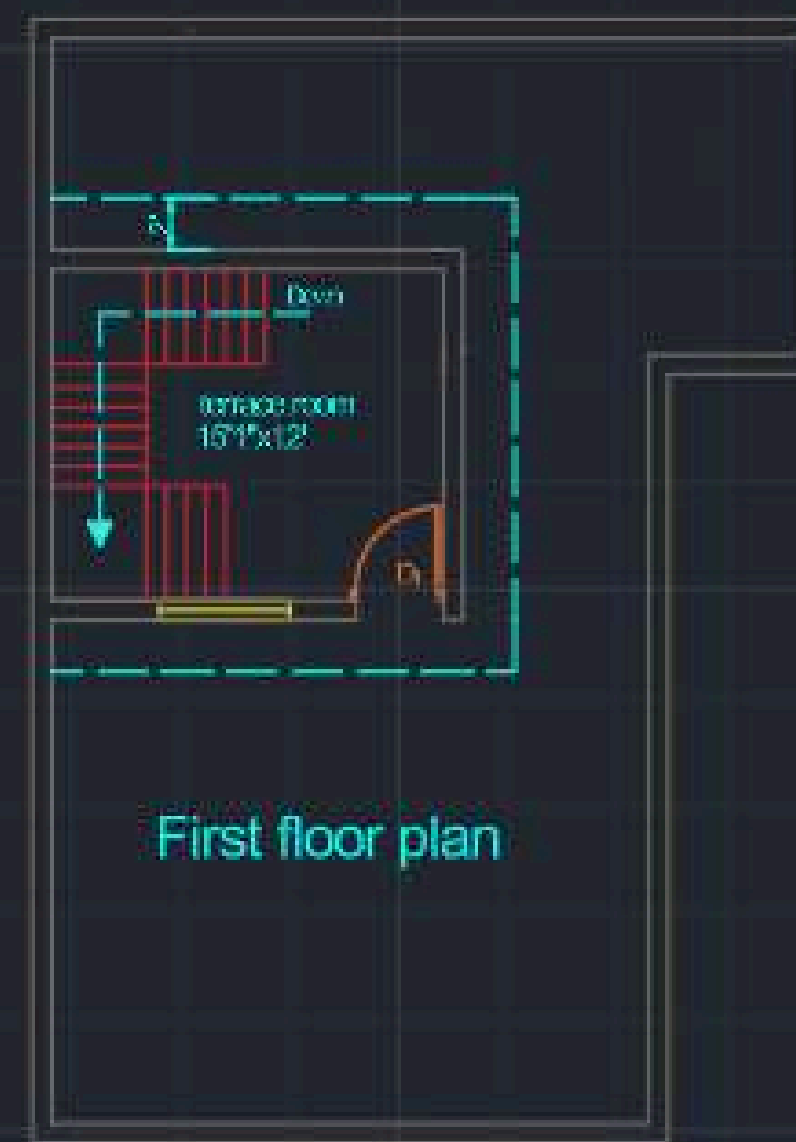
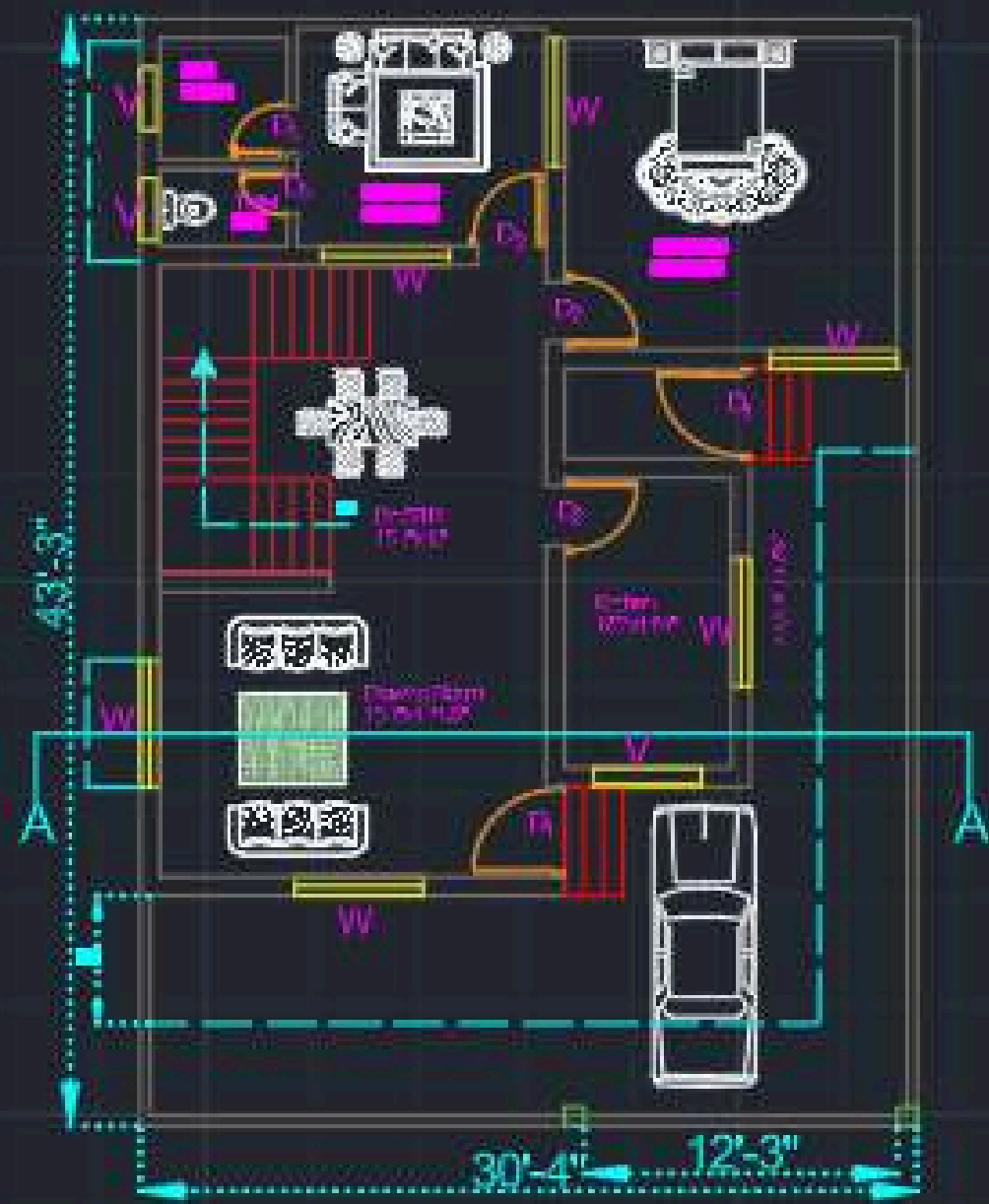
| Name   | ID | Size    |
|--------|----|---------|
| Door   | D1 | 3'5"X9" |
| Door   | D  | 3'X9"   |
| Door   | D2 | 2'6"X9" |
| Window | W3 | 5'X9"   |
| Vent   | V  | 4'3"x9" |

2D Drafting



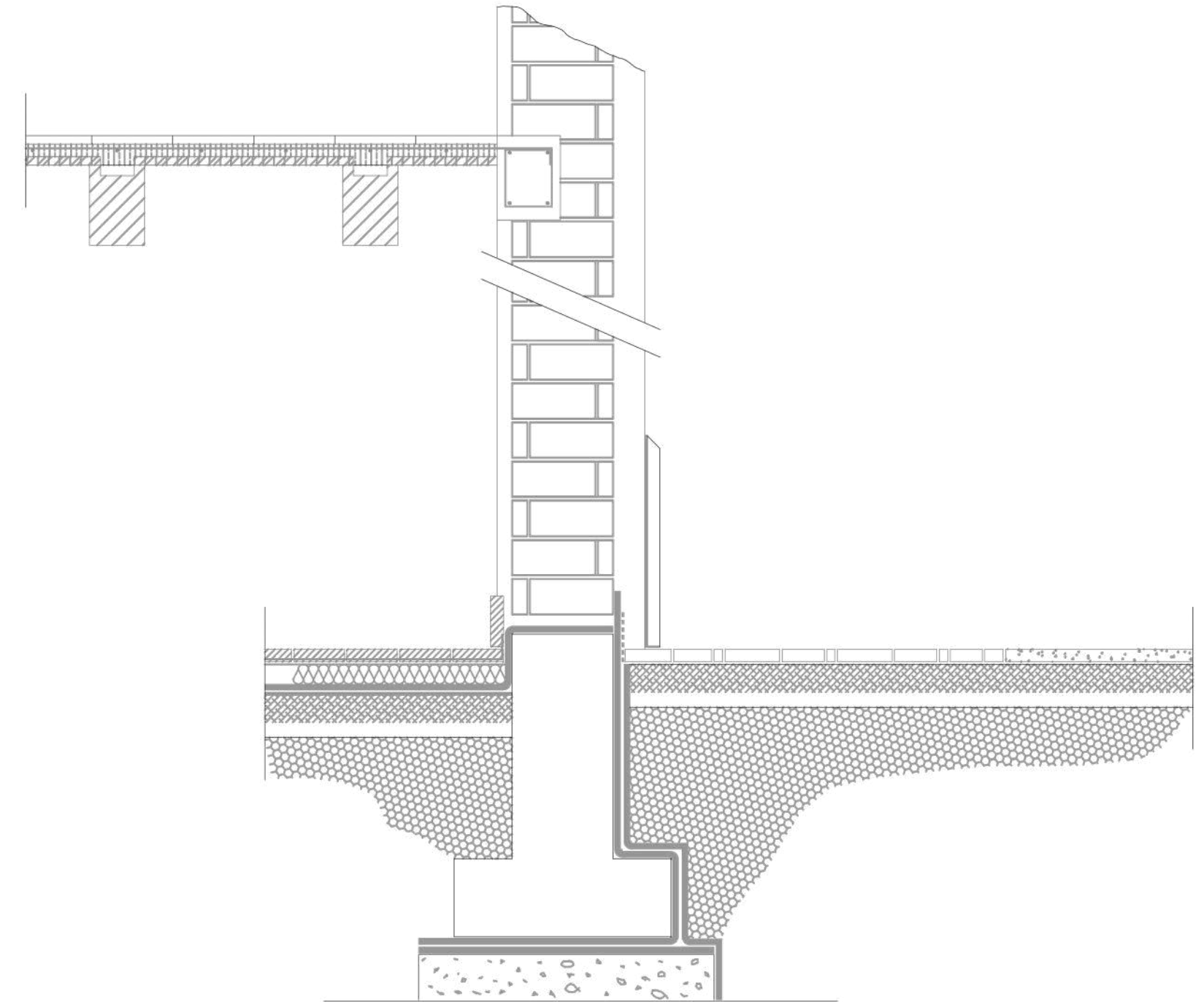
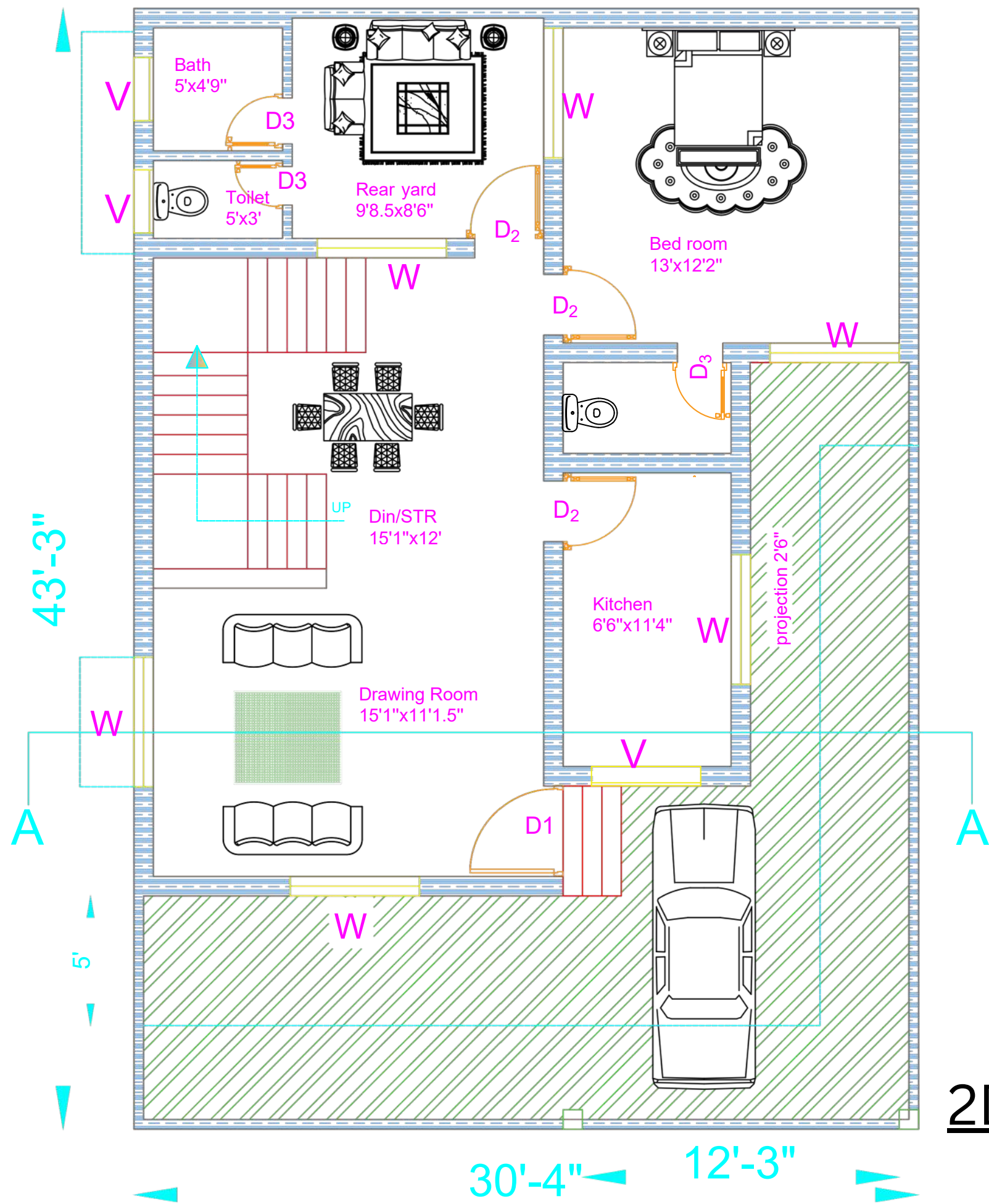
3D MODEL





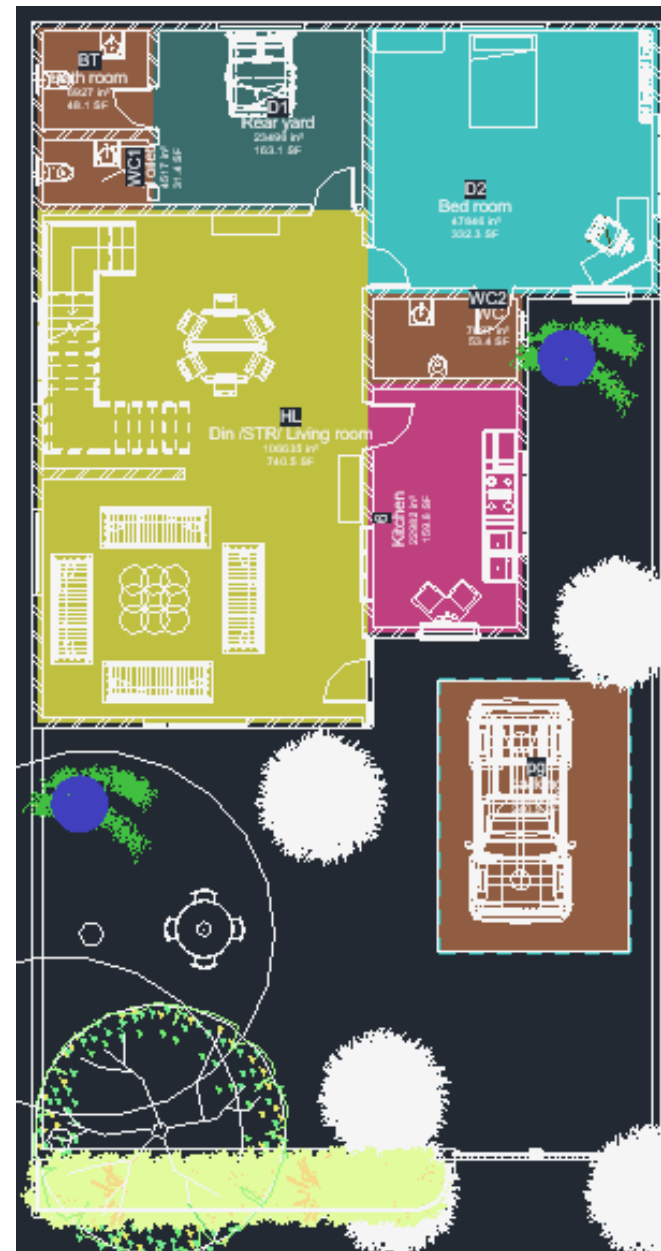
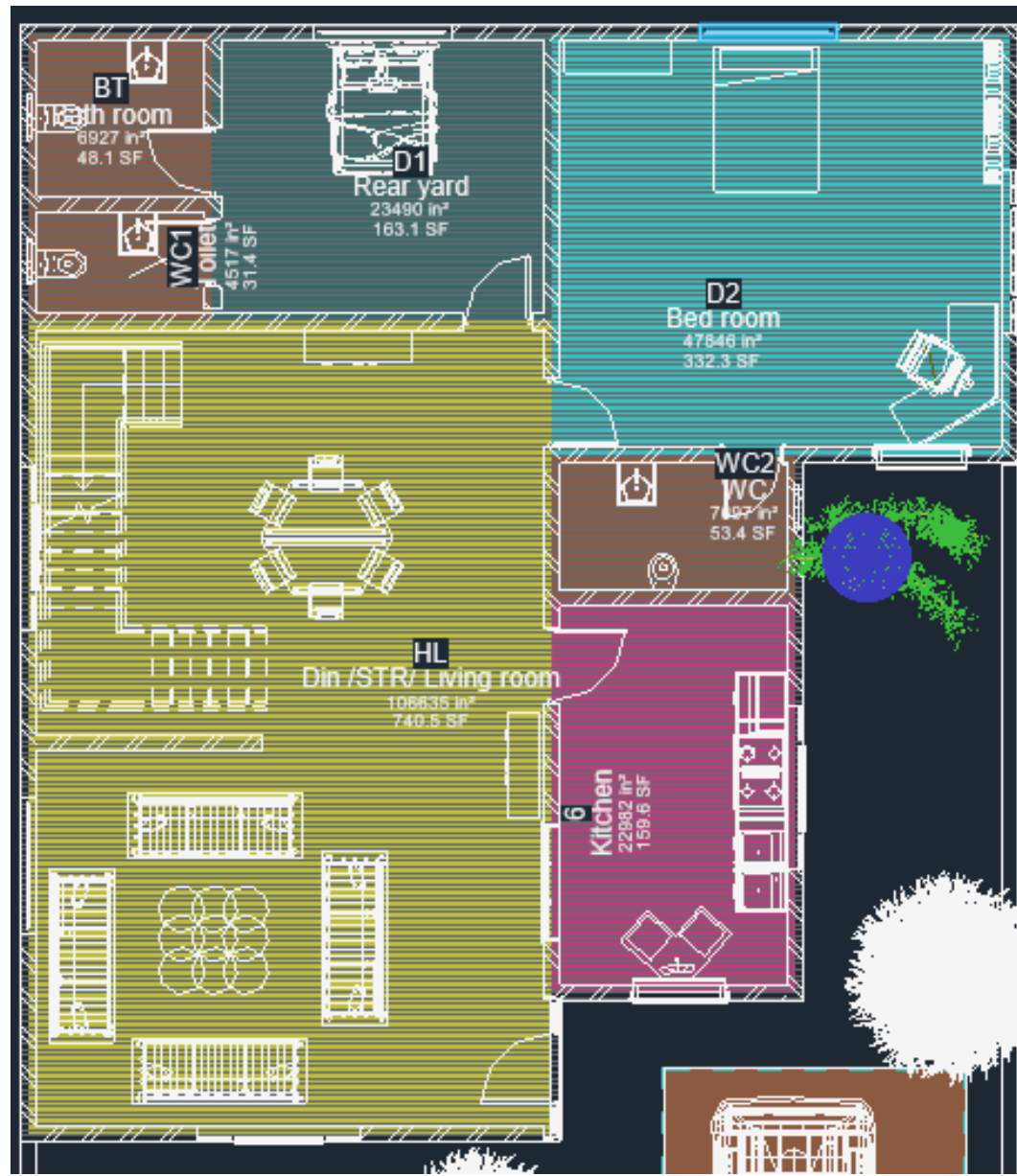
Schematic Section A-A

Schematic Front Elevation



2D Foundation  
 Unites in Feet's  
2D Top View





## By Room Name Legend

- Bath room
- Bed room
- Din /STR/ Living room
- Kitchen
- parking
- Rear yard
- Room
- Toilet
- WC

Top View with Colours

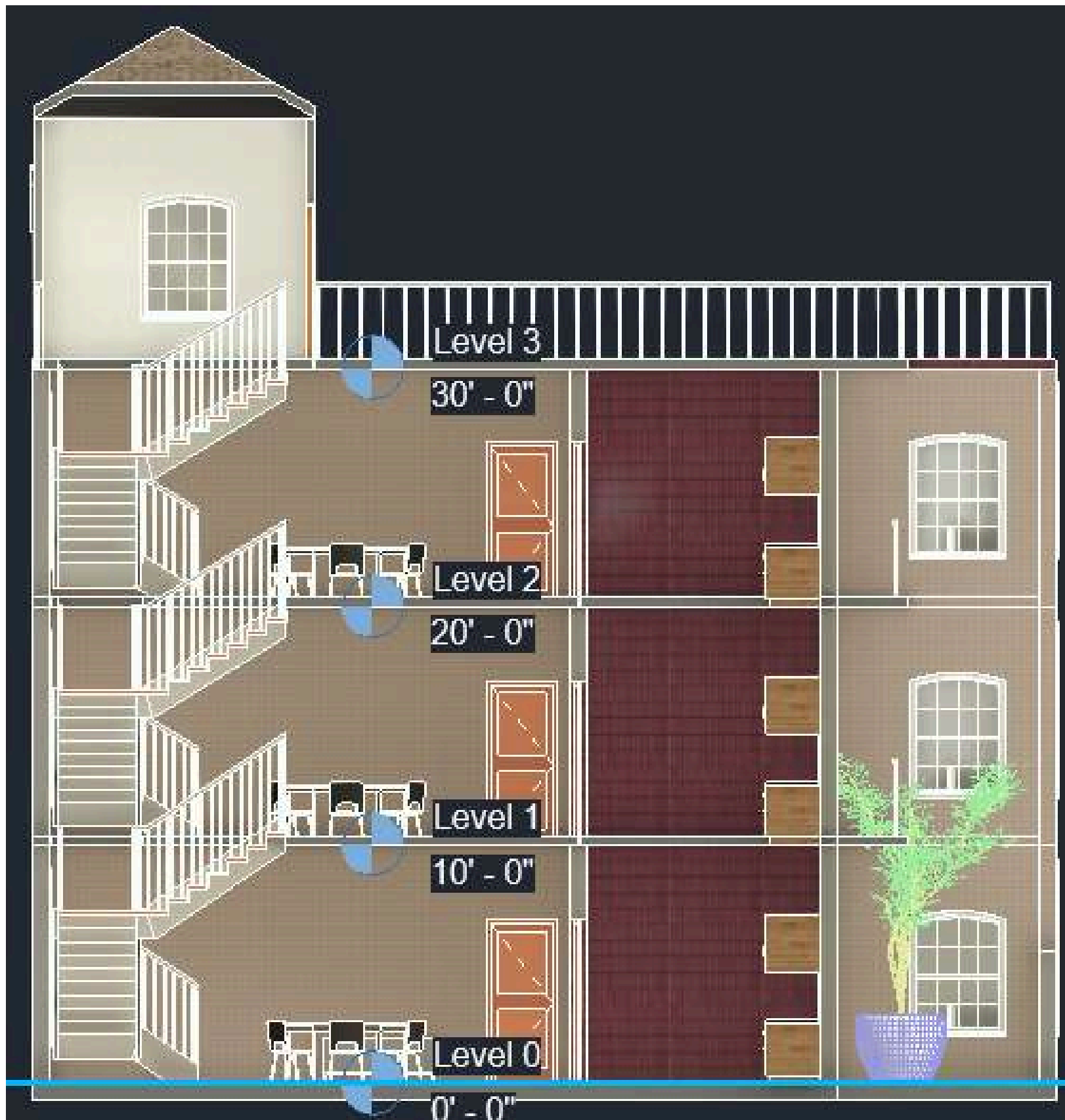




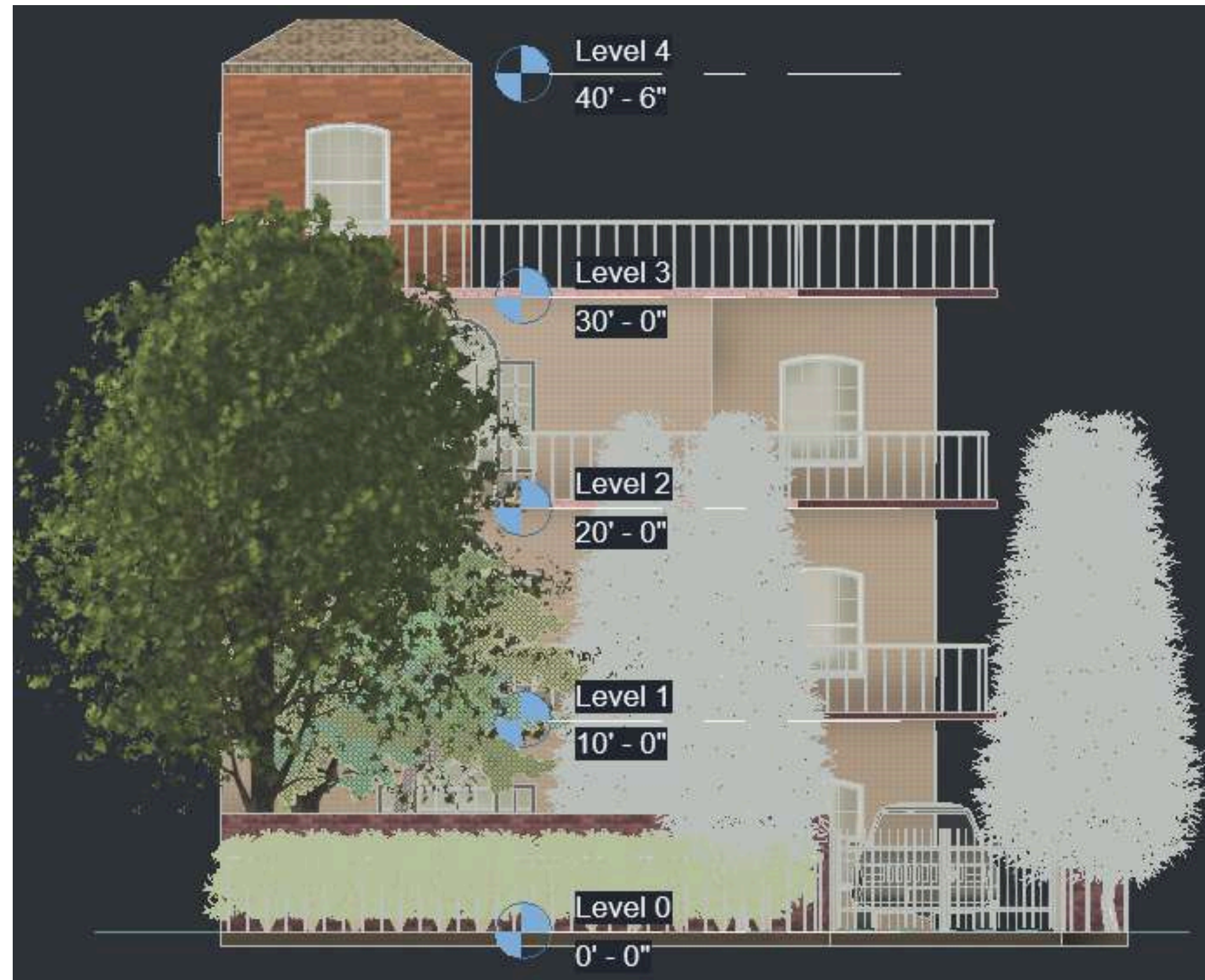
Side Views







cutting of staircase View



Front View



3D View



# Schedule of Door and Window

| Name   | ID | Size    |
|--------|----|---------|
| Door   | D1 | 3'5"X9" |
| Door   | D  | 3'X9"   |
| Door   | D2 | 2'6"X9" |
| Window | W3 | 5'X9"   |
| Vent   | V  | 4'3"x9" |

Basic Parametrises of dimensions

| Area                  | Volume              | Width   | no of brike | cement   | sand     | cost of sand | cost of brikes | cost of cement |
|-----------------------|---------------------|---------|-------------|----------|----------|--------------|----------------|----------------|
| 12202 in <sup>2</sup> | 1.68 m <sup>3</sup> | 0' - 6" | 842.4273    | 2.08922  | 13.14187 | 657.093257   | 8424.27253     | 835.687835     |
| 14398 in <sup>2</sup> | 2.00 m <sup>3</sup> | 0' - 6" | 998.5807    | 2.47648  | 15.57786 | 778.892922   | 9985.806688    | 990.592023     |
| 21792 in <sup>2</sup> | 3.02 m <sup>3</sup> | 0' - 6" | 1511.412    | 3.748301 | 23.57802 | 1178.90112   | 15114.11693    | 1499.320399    |
| 26636 in <sup>2</sup> | 3.63 m <sup>3</sup> | 0' - 6" | 1812.795    | 4.495731 | 28.2796  | 1413.980023  | 18127.94901    | 1798.292542    |
| 2386 in <sup>2</sup>  | 0.33 m <sup>3</sup> | 0' - 6" | 165.4498    | 0.410316 | 2.581018 | 129.050879   | 1654.49845     | 164.126246     |
| 13920 in <sup>2</sup> | 1.93 m <sup>3</sup> | 0' - 6" | 965.4174    | 2.394235 | 15.06051 | 753.025591   | 9654.17424     | 957.694085     |
| 21595 in <sup>2</sup> | 3.00 m <sup>3</sup> | 0' - 6" | 1497.748    | 3.714416 | 23.36488 | 1168.243805  | 14977.48468    | 1485.76648     |
| 50135 in <sup>2</sup> | 6.95 m <sup>3</sup> | 0' - 6" | 3477.171    | 8.623384 | 54.24387 | 2712.193249  | 34771.70832    | 3449.353465    |
| 62050 in <sup>2</sup> | 8.61 m <sup>3</sup> | 0' - 6" | 4303.488    | 10.67265 | 67.13442 | 3356.720749  | 43034.88139    | 4269.060234    |
| 10793 in <sup>2</sup> | 1.50 m <sup>3</sup> | 0' - 6" | 748.5198    | 1.856329 | 11.67691 | 583.84542    | 7485.197695    | 742.531611     |
| 10367 in <sup>2</sup> | 1.44 m <sup>3</sup> | 0' - 6" | 719.0154    | 1.783158 | 11.21664 | 560.831977   | 7190.153555    | 713.263233     |
| 28456 in <sup>2</sup> | 3.95 m <sup>3</sup> | 0' - 6" | 1973.537    | 4.894371 | 30.78717 | 1539.358532  | 19735.3658     | 1957.748287    |
| 36207 in <sup>2</sup> | 5.02 m <sup>3</sup> | 0' - 6" | 2511.094    | 6.227513 | 39.17307 | 1958.653435  | 25110.94148    | 2491.005394    |
| 14104 in <sup>2</sup> | 1.96 m <sup>3</sup> | 0' - 6" | 978.1962    | 2.425926 | 15.25986 | 762.993008   | 9781.96164     | 970.370595     |
| 27193 in <sup>2</sup> | 3.77 m <sup>3</sup> | 0' - 6" | 1885.975    | 4.677217 | 29.4212  | 1471.060162  | 18859.74566    | 1870.88677     |
| 12202 in <sup>2</sup> | 1.68 m <sup>3</sup> | 0' - 6" | 842.4273    | 2.08922  | 13.14187 | 657.093257   | 8424.27253     | 835.687835     |
| 14398 in <sup>2</sup> | 2.00 m <sup>3</sup> | 0' - 6" | 998.5807    | 2.47648  | 15.57786 | 778.892922   | 9985.806688    | 990.592023     |
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| 26636 in <sup>2</sup> | 3.63 m <sup>3</sup> | 0' - 6" | 1812.795    | 4.495731 | 28.2796  | 1413.980023  | 18127.94901    | 1798.292542    |
| 2386 in <sup>2</sup>  | 0.33 m <sup>3</sup> | 0' - 6" | 165.4498    | 0.410316 | 2.581018 | 129.050879   | 1654.49845     | 164.126246     |
| 13920 in <sup>2</sup> | 1.93 m <sup>3</sup> | 0' - 6" | 965.4174    | 2.394235 | 15.06051 | 753.025591   | 9654.17424     | 957.694085     |
| 21595 in <sup>2</sup> | 3.00 m <sup>3</sup> | 0' - 6" | 1497.748    | 3.714416 | 23.36488 | 1168.243805  | 14977.48468    | 1485.76648     |
| 14368 in <sup>2</sup> | 1.11 m <sup>3</sup> | 0' - 6" | 556.181     | 1.379329 | 8.676424 | 433.821202   | 5561.810283    | 551.73158      |
| 16398 in <sup>2</sup> | 1.27 m <sup>3</sup> | 0' - 6" | 634.7448    | 1.574167 | 9.902018 | 495.100909   | 6347.447555    | 629.666797     |
| 14368 in <sup>2</sup> | 1.11 m <sup>3</sup> | 0' - 6" | 556.181     | 1.379329 | 8.676424 | 433.821179   | 5561.809992    | 551.731551     |
| 14841 in <sup>2</sup> | 1.15 m <sup>3</sup> | 0' - 6" | 574.481     | 1.424713 | 8.961903 | 448.095166   | 5744.809824    | 569.885135     |



| Area                  | Volume              | Width   | no of brike | cement   | sand     | cost of sand | cost of brikes | cost of cement |
|-----------------------|---------------------|---------|-------------|----------|----------|--------------|----------------|----------------|
| 23145 in <sup>2</sup> | 3.21 m <sup>3</sup> | 0' - 6" | 1605.201    | 3.9809   | 25.04114 | 1252.057151  | 16052.01475    | 1592.359863    |
| 46615 in <sup>2</sup> | 6.47 m <sup>3</sup> | 0' - 6" | 3232.999    | 8.017838 | 50.43479 | 2521.739256  | 32329.99046    | 3207.135054    |
| 50135 in <sup>2</sup> | 6.95 m <sup>3</sup> | 0' - 6" | 3477.171    | 8.623384 | 54.24387 | 2712.193249  | 34771.70832    | 3449.353465    |
| 56729 in <sup>2</sup> | 7.87 m <sup>3</sup> | 0' - 6" | 3934.499    | 9.757558 | 61.37819 | 3068.909487  | 39344.99342    | 3903.023347    |
| 10793 in <sup>2</sup> | 1.50 m <sup>3</sup> | 0' - 6" | 748.5198    | 1.856329 | 11.67691 | 583.84542    | 7485.197695    | 742.531611     |
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| 12202 in <sup>2</sup> | 1.68 m <sup>3</sup> | 0' - 6" | 842.4273    | 2.08922  | 13.14187 | 657.093257   | 8424.27253     | 835.687835     |
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| 21792 in <sup>2</sup> | 3.02 m <sup>3</sup> | 0' - 6" | 1511.412    | 3.748301 | 23.57802 | 1178.90112   | 15114.11693    | 1499.320399    |
| 26636 in <sup>2</sup> | 3.63 m <sup>3</sup> | 0' - 6" | 1812.795    | 4.495731 | 28.2796  | 1413.980023  | 18127.94901    | 1798.292542    |
| 2386 in <sup>2</sup>  | 0.33 m <sup>3</sup> | 0' - 6" | 165.4498    | 0.410316 | 2.581018 | 129.050879   | 1654.49845     | 164.126246     |
| 13920 in <sup>2</sup> | 1.93 m <sup>3</sup> | 0' - 6" | 965.4174    | 2.394235 | 15.06051 | 753.025591   | 9654.17424     | 957.694085     |
| 21595 in <sup>2</sup> | 3.00 m <sup>3</sup> | 0' - 6" | 1497.748    | 3.714416 | 23.36488 | 1168.243805  | 14977.48468    | 1485.76648     |
| 23165 in <sup>2</sup> | 3.21 m <sup>3</sup> | 0' - 6" | 1606.611    | 3.984395 | 25.06313 | 1253.156586  | 16066.11008    | 1593.75812     |
| 2563 in <sup>2</sup>  | 0.36 m <sup>3</sup> | 0' - 6" | 177.7659    | 0.440859 | 2.773148 | 138.65738    | 1777.658717    | 176.343745     |
| 50135 in <sup>2</sup> | 6.95 m <sup>3</sup> | 0' - 6" | 3477.171    | 8.623384 | 54.24387 | 2712.193249  | 34771.70832    | 3449.353465    |
| 56729 in <sup>2</sup> | 7.87 m <sup>3</sup> | 0' - 6" | 3934.499    | 9.757558 | 61.37819 | 3068.909487  | 39344.99342    | 3903.023347    |
| 10793 in <sup>2</sup> | 1.50 m <sup>3</sup> | 0' - 6" | 748.5198    | 1.856329 | 11.67691 | 583.84542    | 7485.197695    | 742.531611     |
| 10367 in <sup>2</sup> | 1.44 m <sup>3</sup> | 0' - 6" | 719.0154    | 1.783158 | 11.21664 | 560.831977   | 7190.153555    | 713.263233     |
| 28456 in <sup>2</sup> | 3.95 m <sup>3</sup> | 0' - 6" | 1973.537    | 4.894371 | 30.78717 | 1539.358532  | 19735.3658     | 1957.748287    |
| 36207 in <sup>2</sup> | 5.02 m <sup>3</sup> | 0' - 6" | 2511.094    | 6.227513 | 39.17307 | 1958.653435  | 25110.94148    | 2491.005394    |
| 14104 in <sup>2</sup> | 1.96 m <sup>3</sup> | 0' - 6" | 978.1962    | 2.425926 | 15.25986 | 762.993008   | 9781.96164     | 970.370595     |
| 27193 in <sup>2</sup> | 3.77 m <sup>3</sup> | 0' - 6" | 1885.975    | 4.677217 | 29.4212  | 1471.060162  | 18859.74566    | 1870.88677     |

# 2BHK Home Design

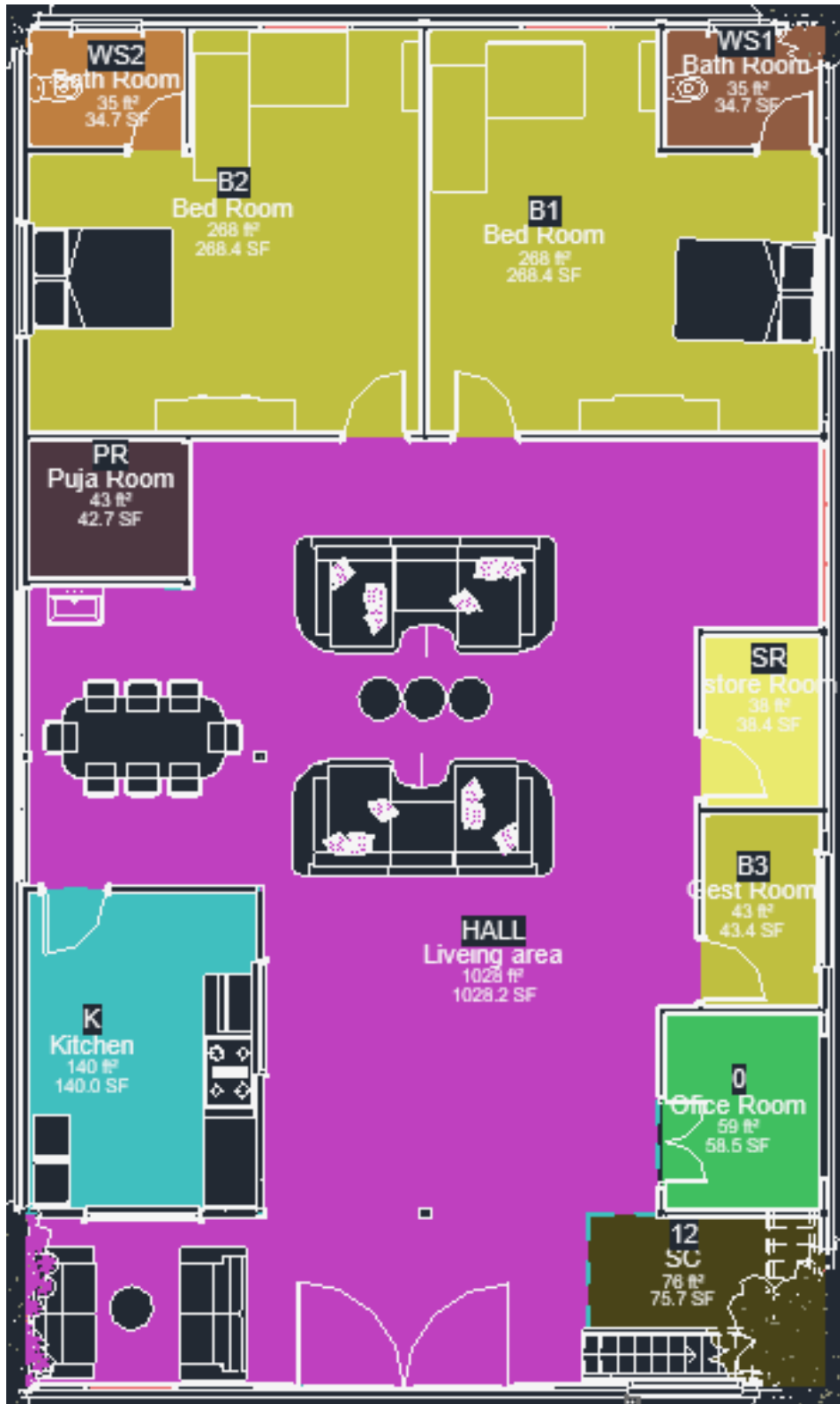
- The plan shows a well-organized 2BHK house with two large bedrooms, a spacious living hall, kitchen, bathrooms, and additional rooms like a puja room, guest room, office room, and store room. The zoning is properly divided into private spaces (bedrooms) and common areas (hall & kitchen).







3D VIEW



TOP VIEW

## By Room Number Legend

|    |      |     |
|----|------|-----|
| 0  | B3   | PR  |
| 12 | HALL | SR  |
| B1 | K    | WS1 |
| B2 | P    | WS2 |

## List of Colours with Colour Icons

- Emerald Green – Office Room (O)
- Olive Brown – W.C (12)
- Pastel Lime – Bedroom 1 (B1)
- Chartreuse Green – Bedroom 2 (B2)
- Olive Yellow – Guest Room (B3)
- Magenta / Orchid Purple – Hall (Living Area)
- Turquoise / Aqua Blue – Kitchen (K)
- Sage Green – Passage (P)
- Plum Purple – Puja Room (PR)
- Light Lemon Yellow – Store Room (SR)
- Terracotta Brown – Bath Room (WS1)
- Tan / Light Terracotta – Bath Room (WS2)







BACK VIEW

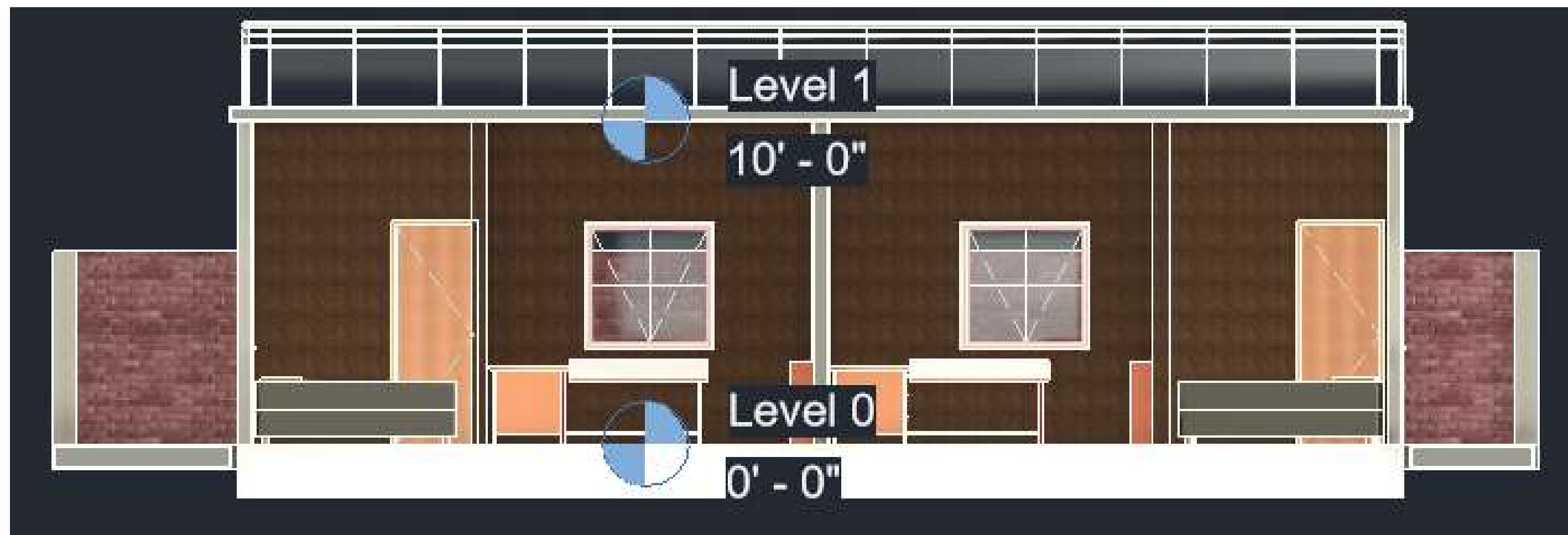
FRONR VIEW

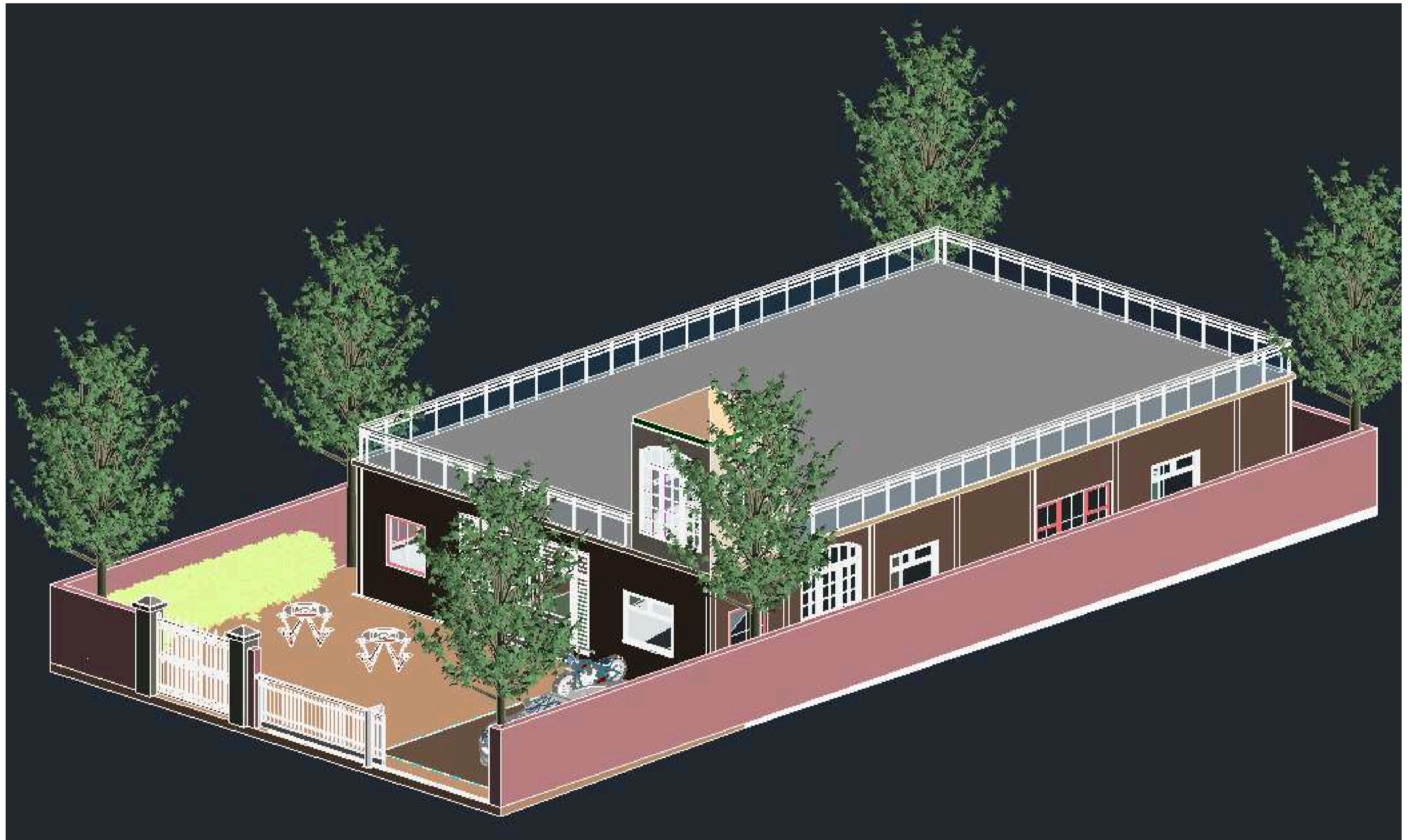




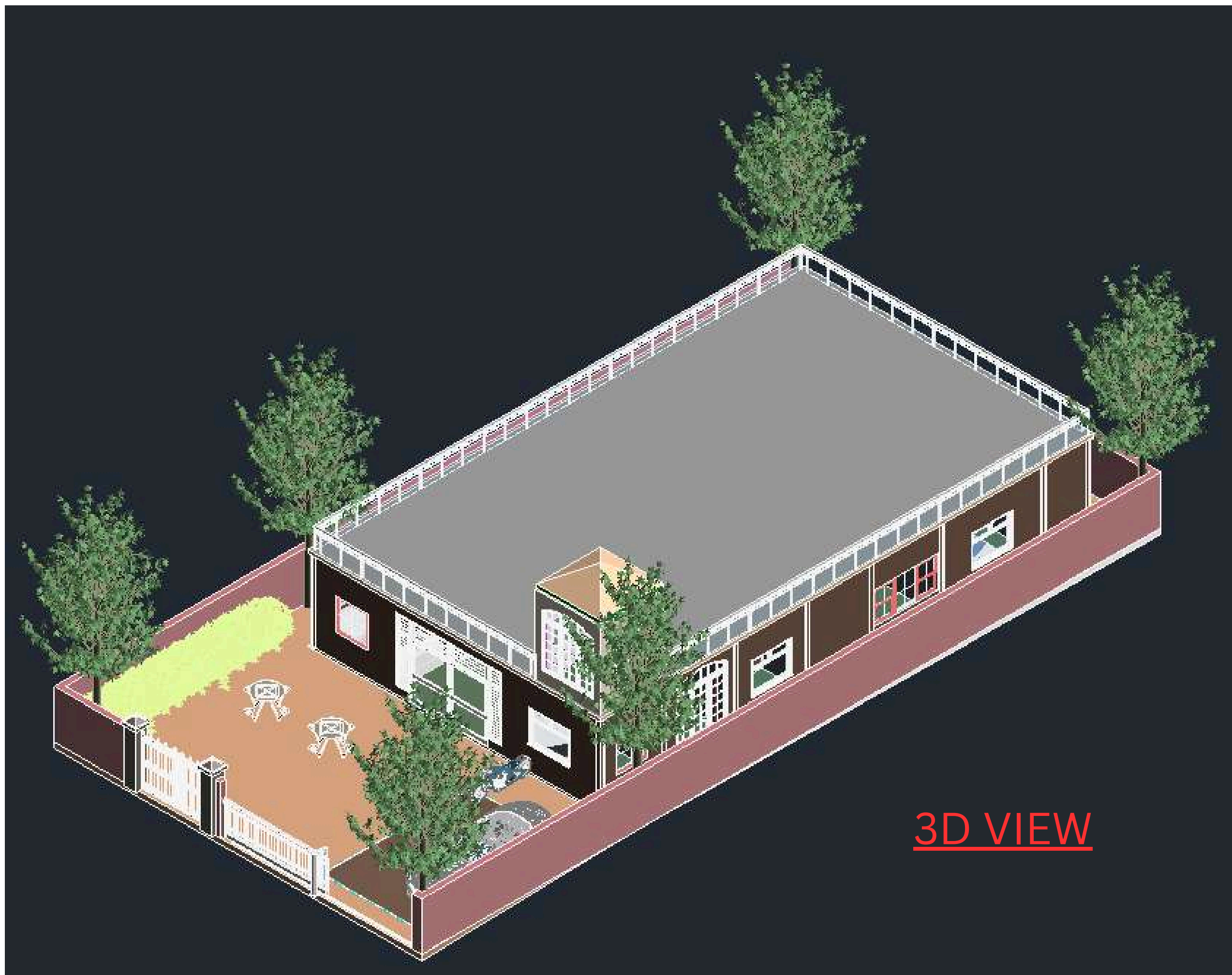


SECTION VIEWS







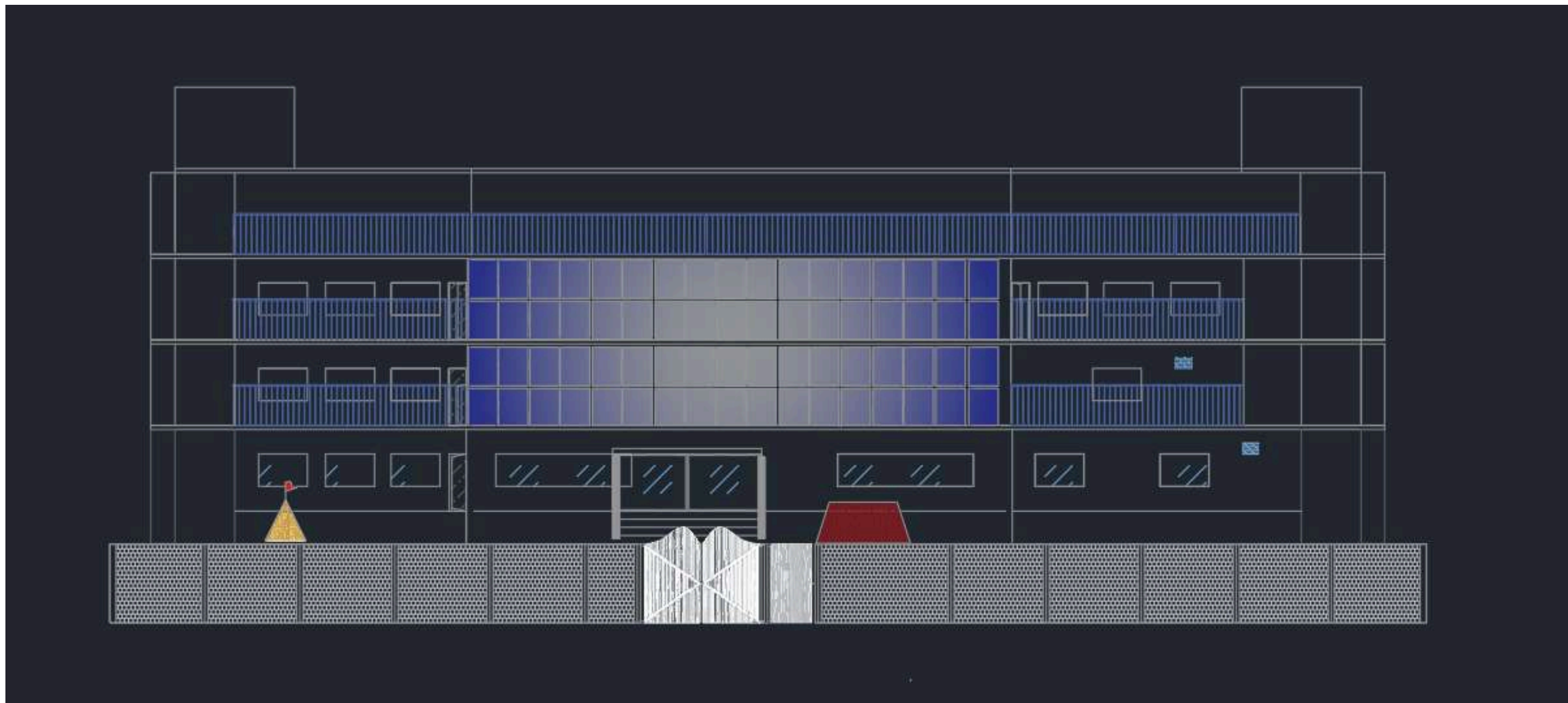


3D VIEW

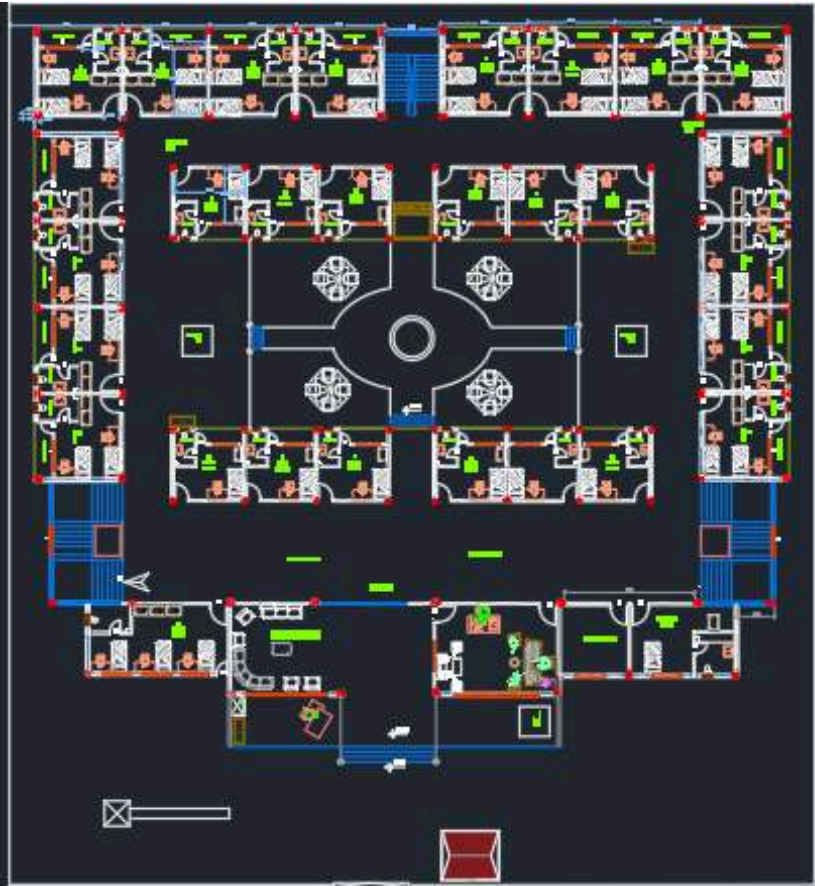
## 4)Topaz Hostel

- Implementing BIM for the Topaz Hostel transforms the design from a static 2D drawing into a dynamic digital archive. By modeling single beds as Parametric Families, Revit stores vital metadata—such as cost and maintenance schedules—allowing for global updates across the repetitive hostel layout with a single change in the Family Editor. These intelligent objects also automate space analysis, ensuring all rooms maintain code-compliant clearance zones.
- The balcony is archived as a critical structural extension, where Revit tracks load-bearing capacity and thermal performance. Safety is managed through Railing Families that automatically adjust to balcony dimensions, maintaining permanent code compliance within the model.
- Regarding the Time Complication (4D BIM), the workflow requires a "front-loaded" effort, demanding more initial time than 2D CAD to define complex layers. However, this prevents costly on-site delays by enabling Conflict Resolution, where Revit flags "clashes"—such as a balcony column interfering with a bed—during the design phase. Ultimately, this high-fidelity archive ensures a "just-in-time" delivery schedule, turning initial modeling complexity into long-term construction insurance.

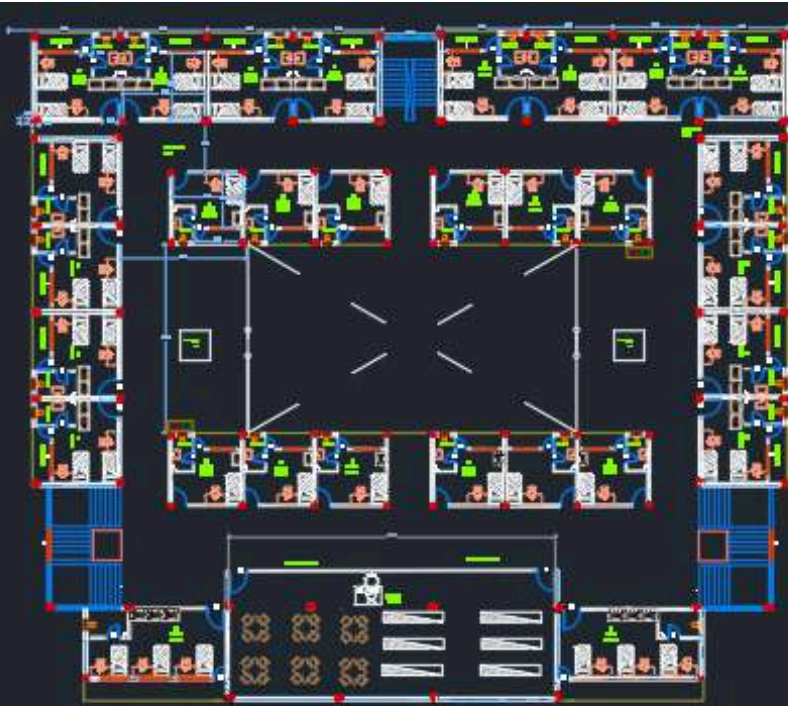




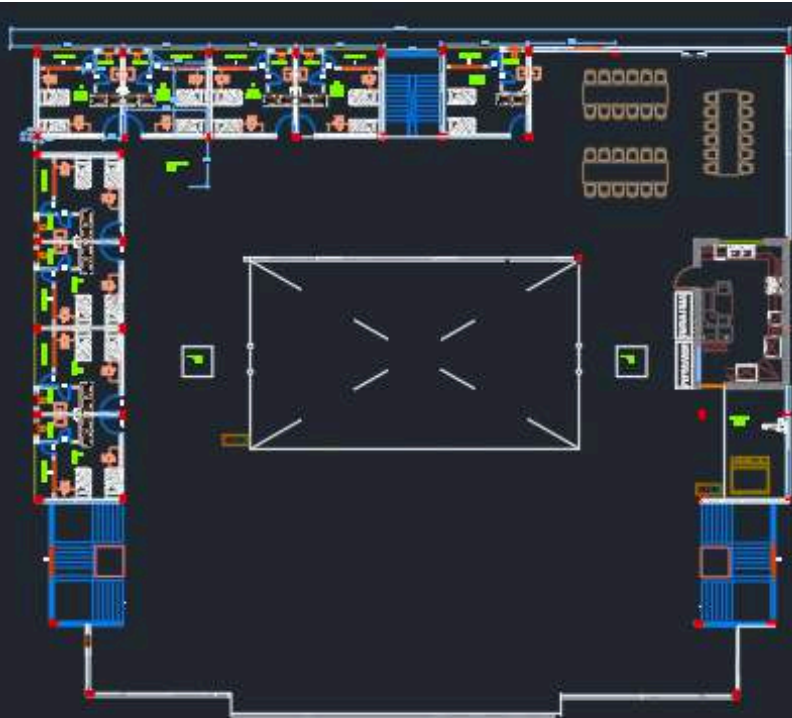
2D Drafting



1st FLOOR



2nd FLOOR



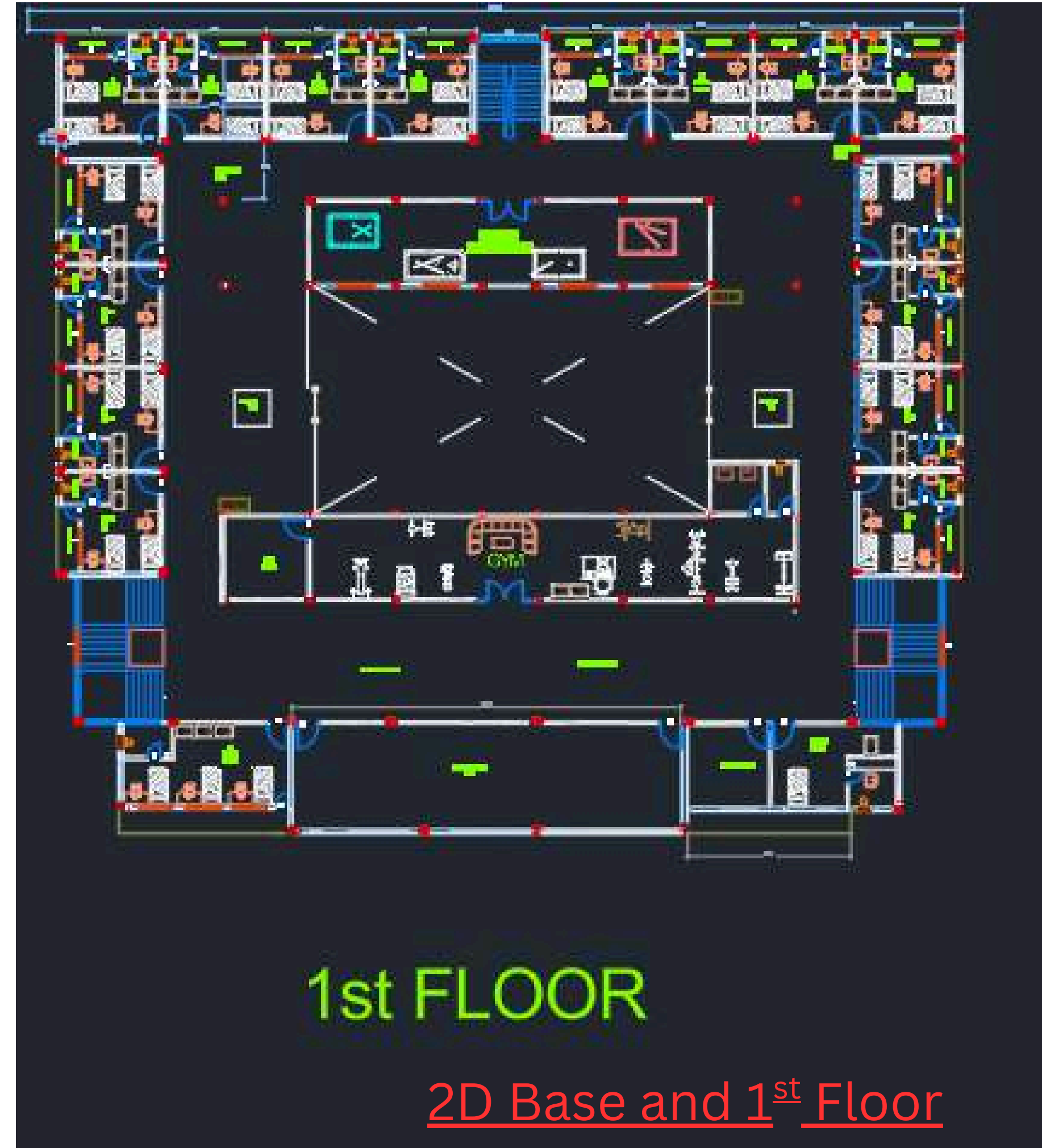
3rd FLOOR





3D MODEL





A 3D architectural rendering of the front view of the Topaz Hostel. The building is a three-story structure with a light beige facade and a grid-like window pattern. It features three white, conical-roofed towers on the roof. A large, leafy green tree stands to the left of the building. In the foreground, there is a dark red brick wall with a white metal gate. The text "Topaz Hostel" is displayed on the left side of the wall. The sky is a clear, bright blue.

Topaz Hostel

front View





Side Views





Top Views





Back View





Top View





cutting of staircase View



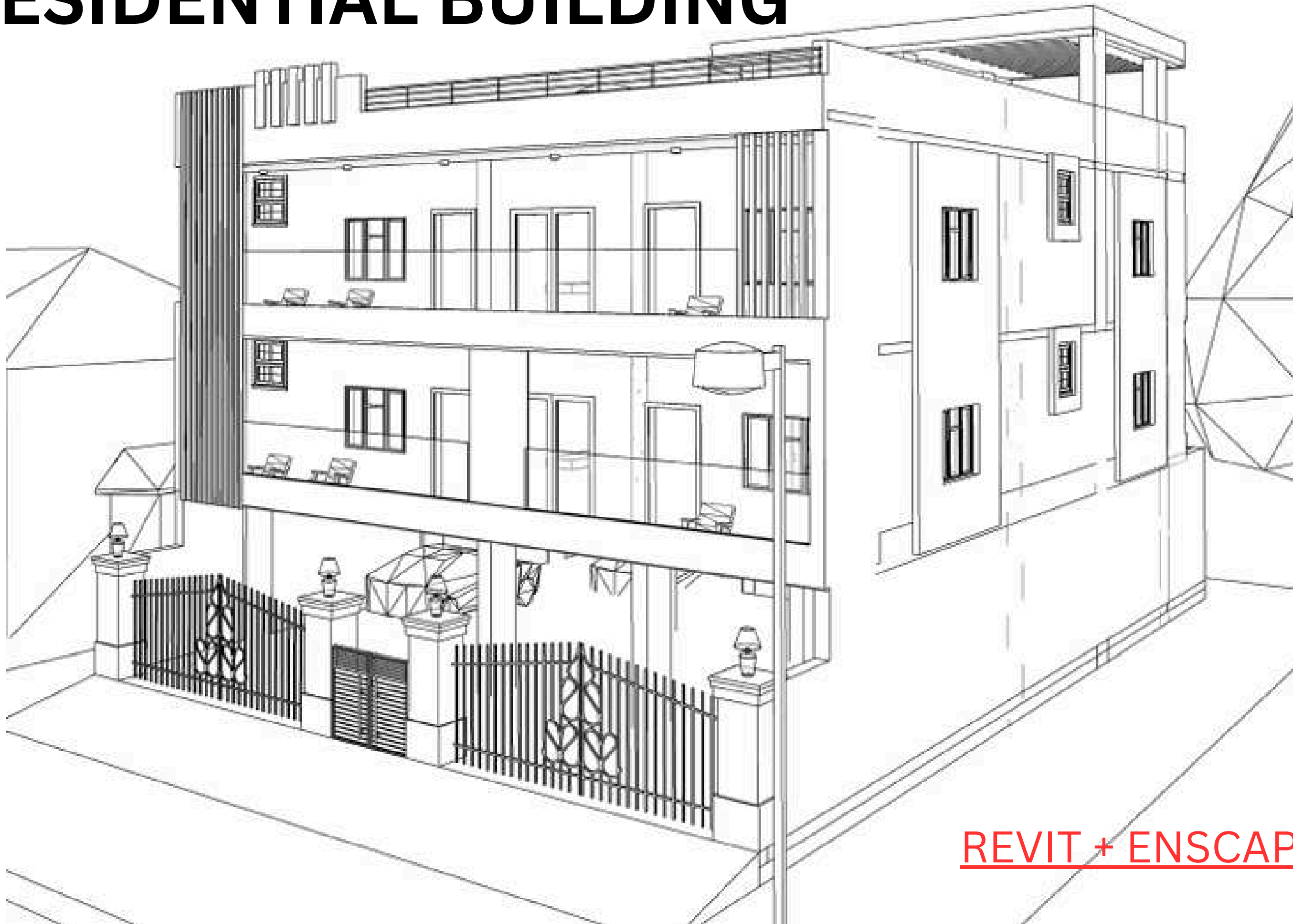






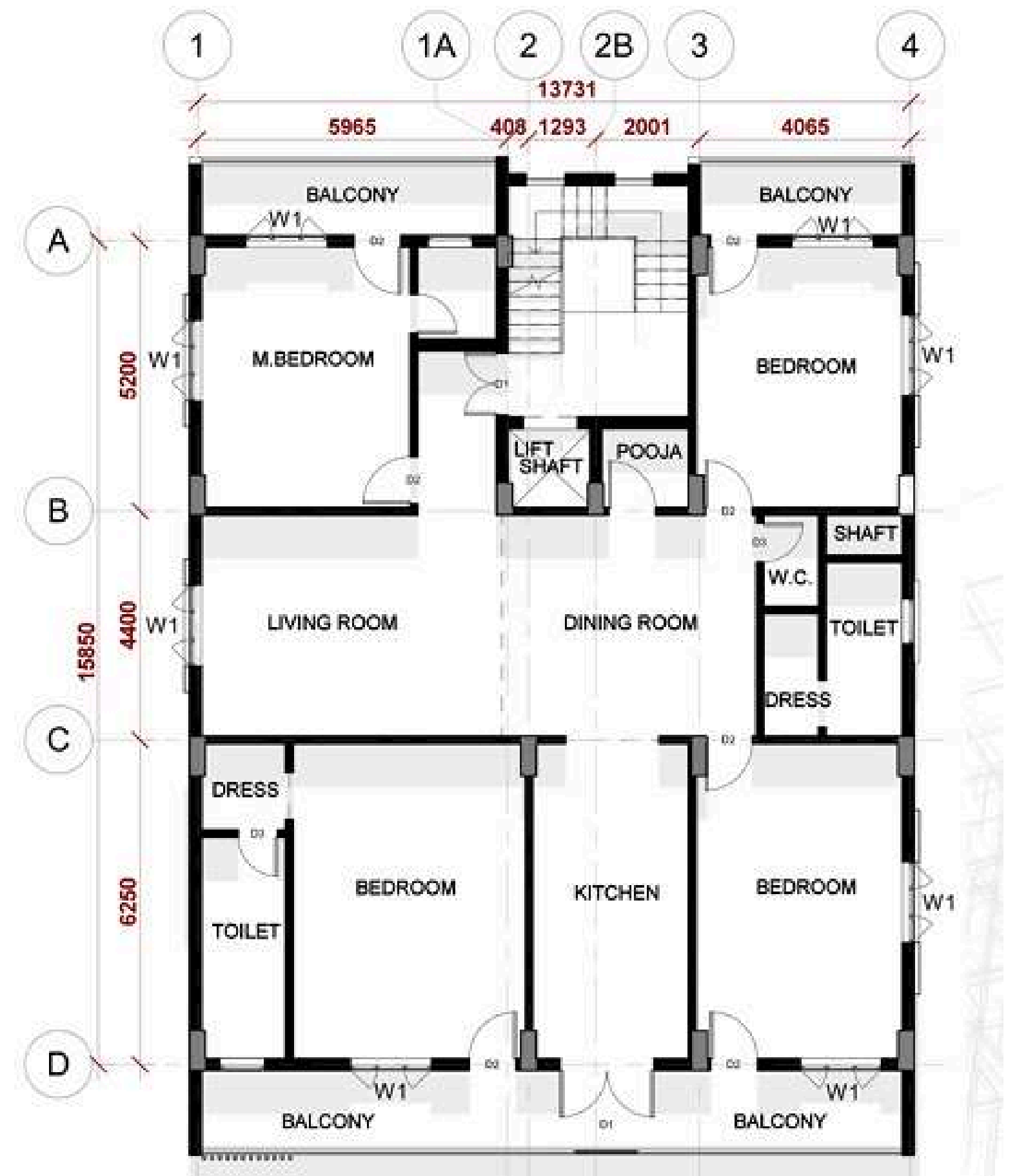
3D MODEL

# G+2 RESIDENTIAL BUILDING

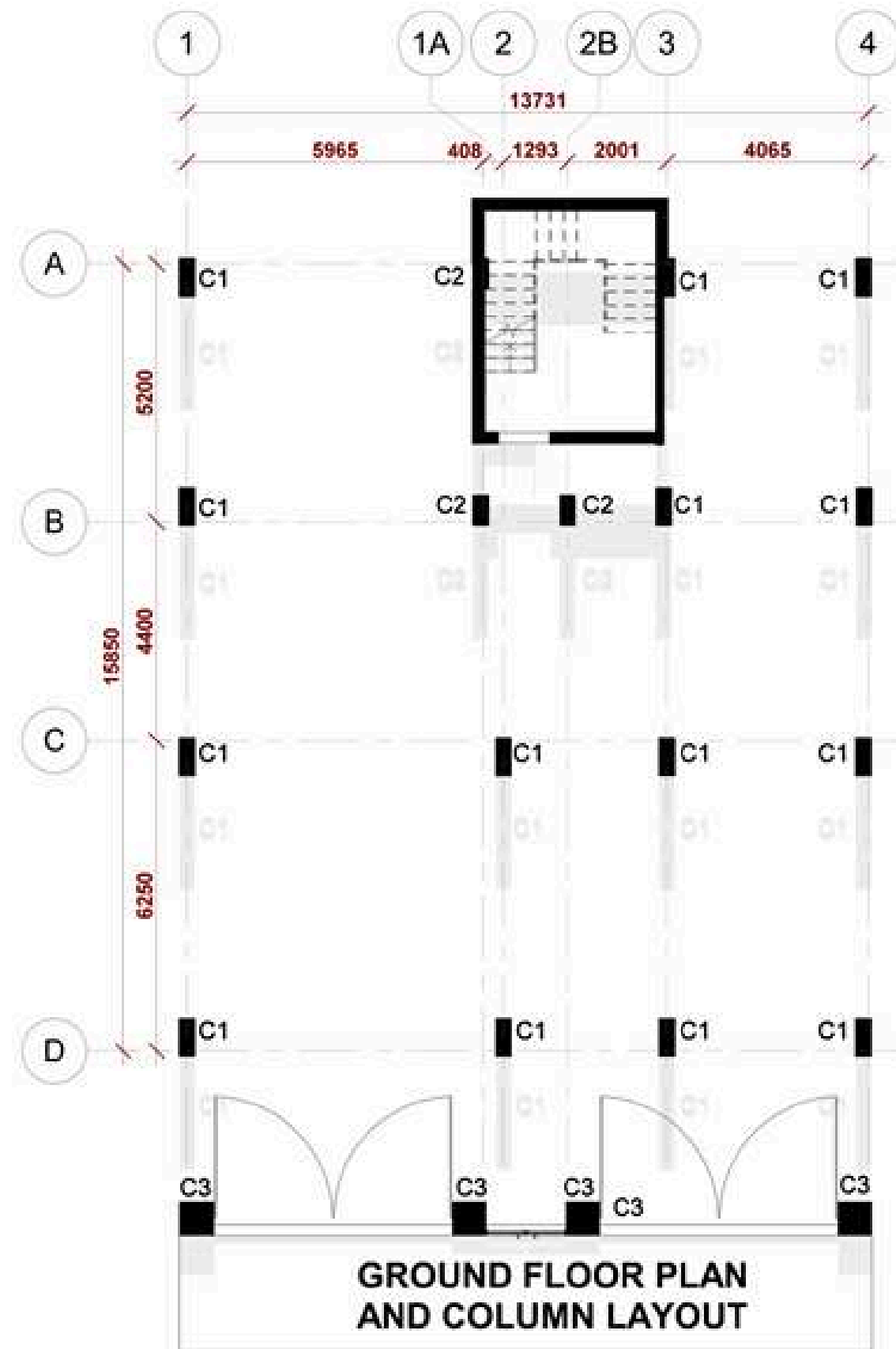


REVIT + ENSCAPE





1,2 FLOOR PLAN



**Colour Filled Legends**

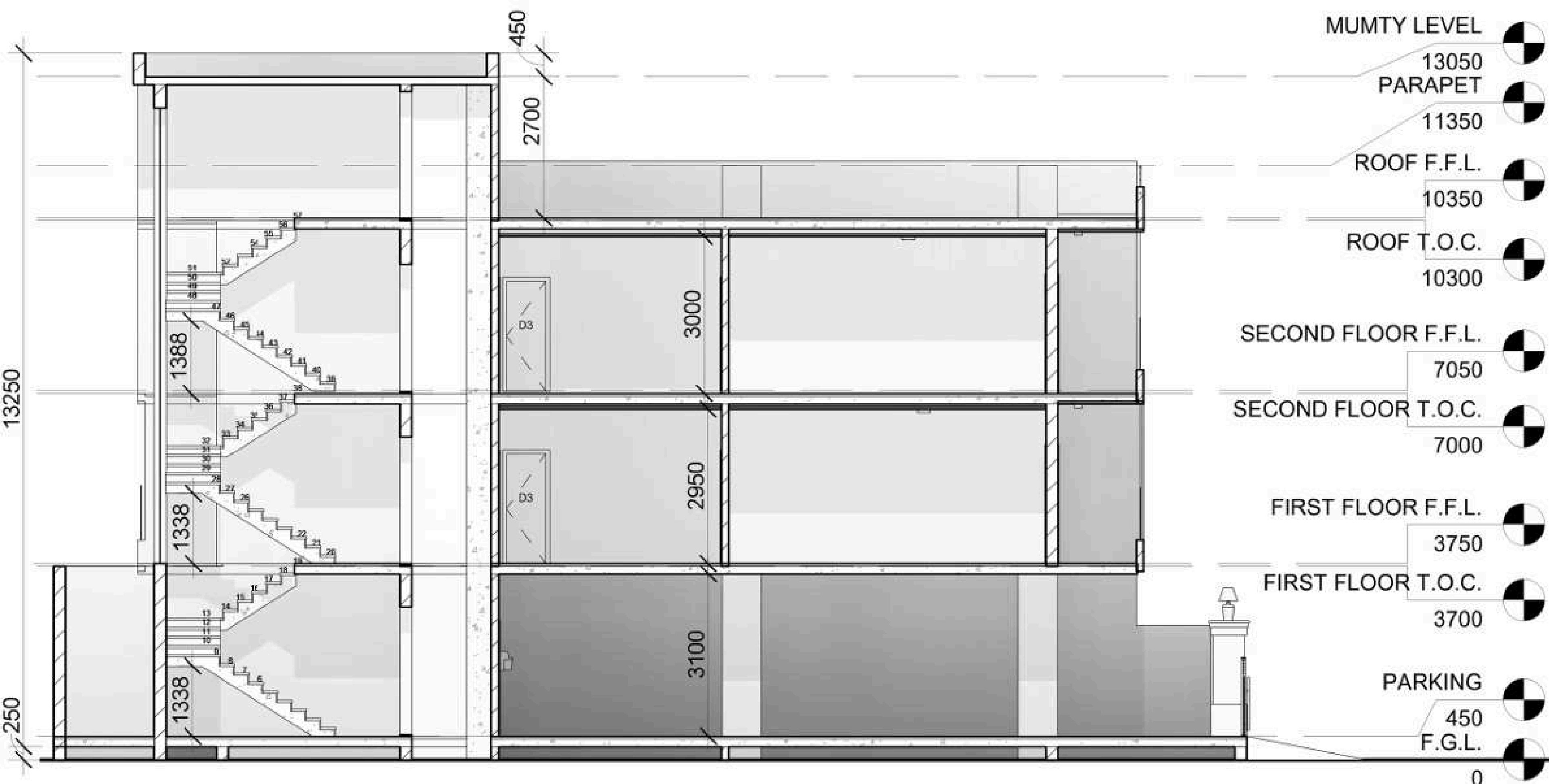


**Room Legend**

- BALCONY
- BEDROOM-1
- BEDROOM-2
- BEDROOM-3
- DINING ROOM
- DRESS
- KITCHEN
- LIFT SHAFT
- LIVING ROOM
- M.BEDROOM
- POOJA
- SHAFT
- STAIRCASE
- TOILET-1
- TOILET-2
- TOILET-3
- W.C.

**FLOOR PLANS**





A-A SECTION VIEW







3D RENDER VIEW



Side Angle 3D View



Back View

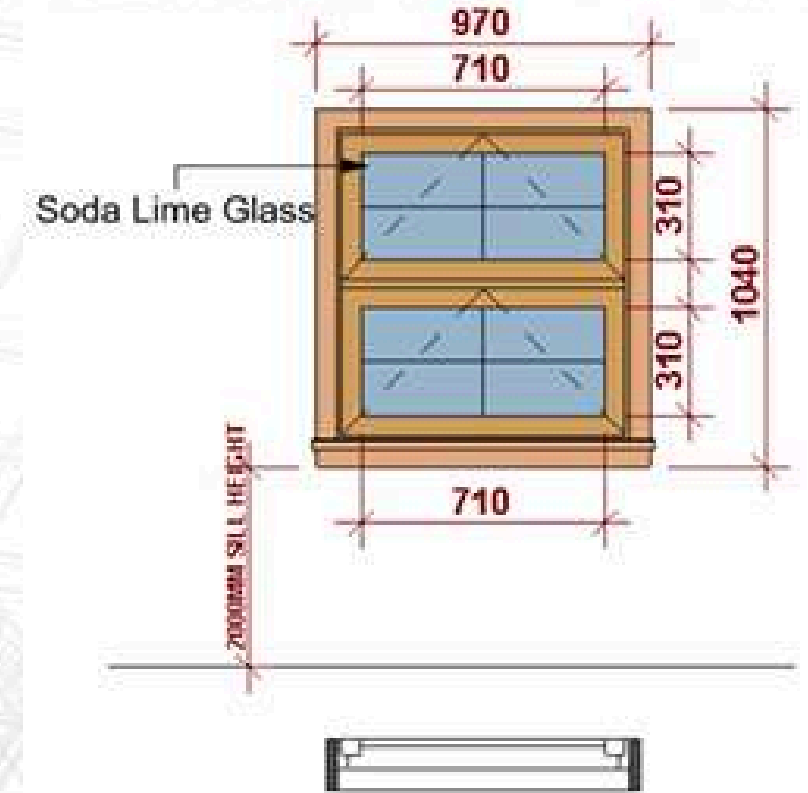
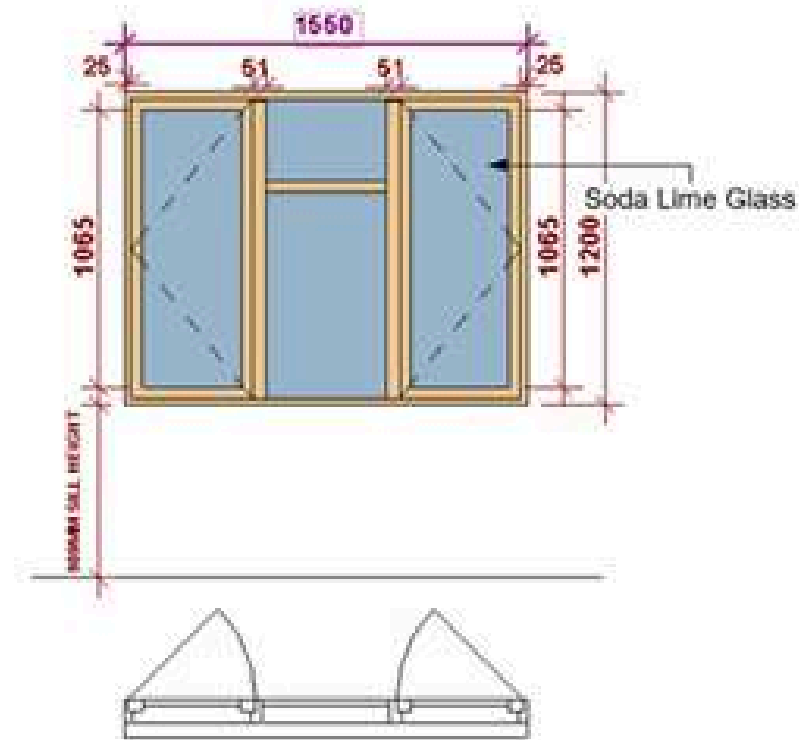
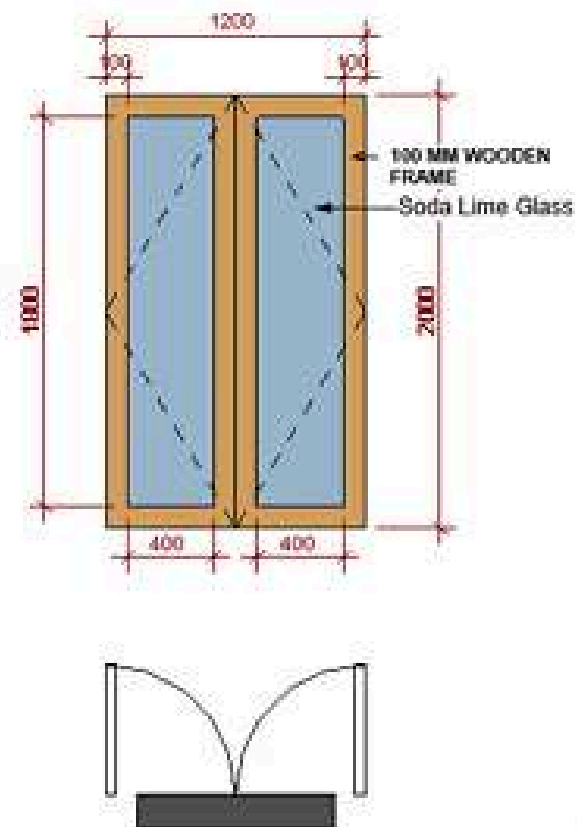


Night View 1



Night View 2





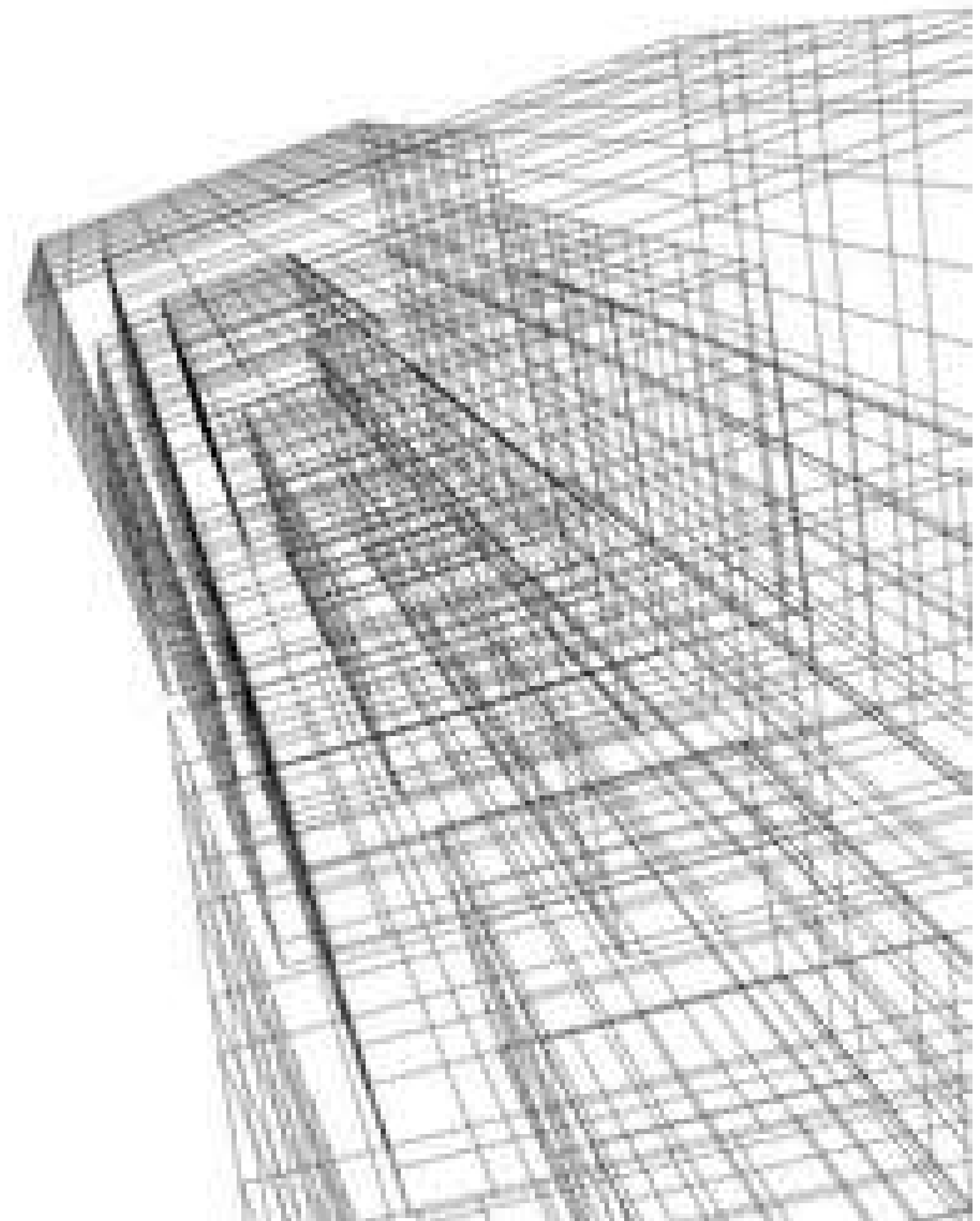
### <Window Schedule>

| A     | B    | C       | D                | E     | F           | G      | H            |
|-------|------|---------|------------------|-------|-------------|--------|--------------|
| Count | Mark | Cost    | Level            | Width | Sill Height | Height | Description  |
|       | 02   | 2500.00 | FIRST FLOOR F.F. | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | FIRST FLOOR F.F. | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | FIRST FLOOR F.F. | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 03   | 2500.00 | FIRST FLOOR F.F. | 750   | 0           | 2100   | WOODEN DOOR  |
|       | 03   | 2500.00 | FIRST FLOOR F.F. | 750   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | FIRST FLOOR F.F. | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | FIRST FLOOR F.F. | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 01   | 4500.00 | FIRST FLOOR F.F. | 1800  | 0           | 2200   | SLIDING DOOR |
|       | 01   | 3000.00 | FIRST FLOOR F.F. | 1200  | 0           | 2000   | WOODEN DOOR  |
|       | 02   | 2500.00 | FIRST FLOOR F.F. | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | FIRST FLOOR F.F. | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | FIRST FLOOR F.F. | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | FIRST FLOOR F.F. | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 03   | 2500.00 | FIRST FLOOR F.F. | 750   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | PARKING          | 900   | 50          | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | PARKING          | 900   | 50          | 2100   | WOODEN DOOR  |
|       | 01   | 3000.00 | PARKING          | 1200  | 50          | 2000   | WOODEN DOOR  |
|       | 02   | 2500.00 | SECOND FLOOR F   | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | SECOND FLOOR F   | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | SECOND FLOOR F   | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 03   | 2500.00 | SECOND FLOOR F   | 750   | 0           | 2100   | WOODEN DOOR  |
|       | 03   | 2500.00 | SECOND FLOOR F   | 750   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | SECOND FLOOR F   | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | SECOND FLOOR F   | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 01   | 3000.00 | SECOND FLOOR F   | 1200  | 0           | 2000   | WOODEN DOOR  |
|       | 02   | 2500.00 | SECOND FLOOR F   | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | SECOND FLOOR F   | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | SECOND FLOOR F   | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 02   | 2500.00 | SECOND FLOOR F   | 900   | 0           | 2100   | WOODEN DOOR  |
|       | 03   | 2500.00 | SECOND FLOOR F   | 750   | 0           | 2100   | WOODEN DOOR  |
|       | 04   | 3000.00 | ROOF F.F.L.      | 1200  | 0           | 2000   | WOODEN DOOR  |
|       | 01   | 4500.00 | SECOND FLOOR T   | 1800  | 0           | 2200   | SLIDING DOOR |

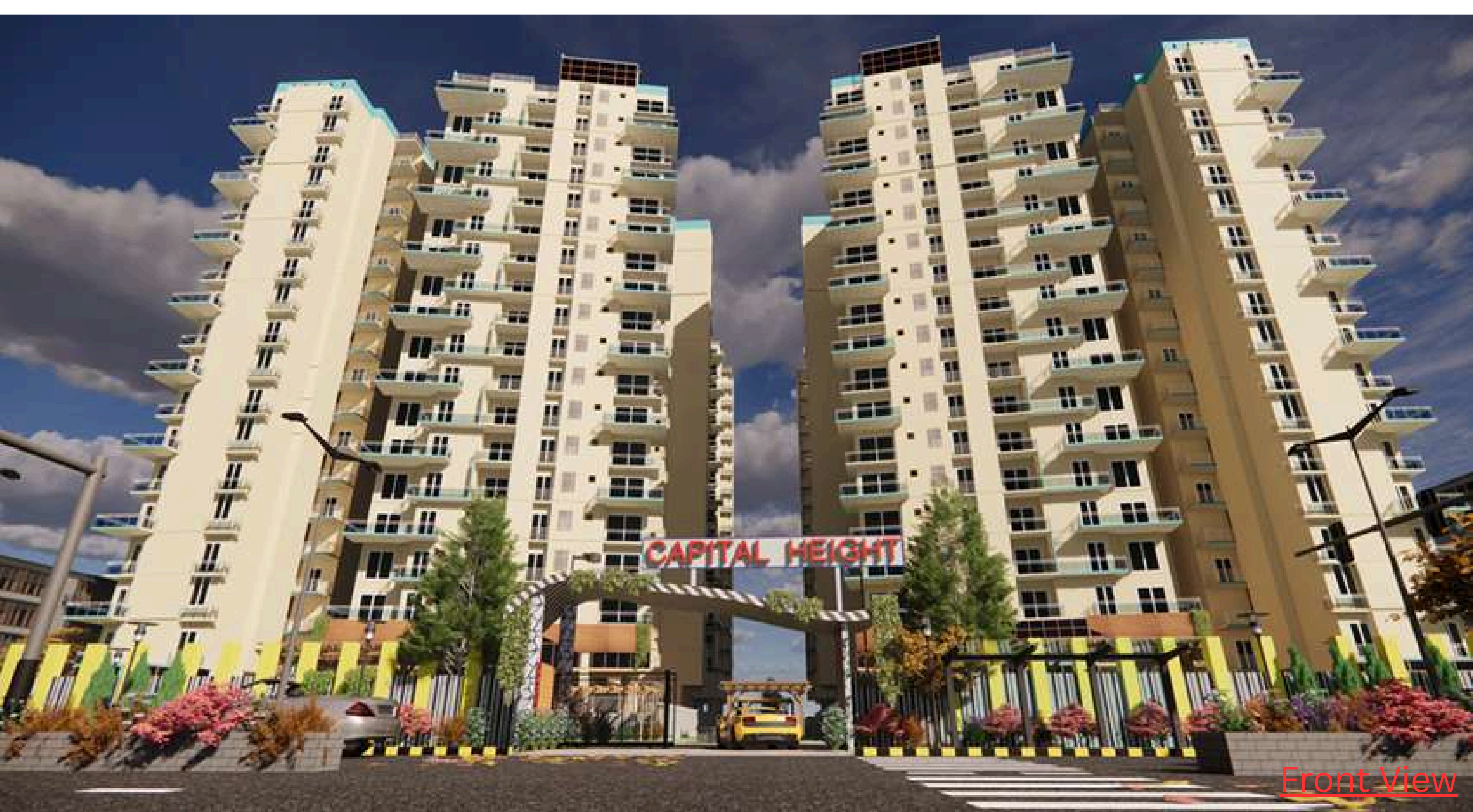
| A     | B    | C       | D                 | E      | F           | G     |
|-------|------|---------|-------------------|--------|-------------|-------|
| Count | Mark | Cost    | Description       | Height | Sill Height | Width |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 976         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 976         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 976         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 976         | 1550  |
| 1     | V1   | 2000.00 | GLAZED VENTILATOR | 900    | 2000        | 850   |
| 1     | V1   | 2000.00 | GLAZED VENTILATOR | 900    | 2000        | 850   |
| 1     | V1   | 2000.00 | GLAZED VENTILATOR | 900    | 2000        | 850   |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | W1   | 4000.00 | GLAZED WINDOW     | 1200   | 900         | 1550  |
| 1     | V1   | 2000.00 | GLAZED VENTILATOR | 900    | 2000        | 850   |
| 1     | V1   | 2000.00 | GLAZED VENTILATOR | 900    | 2000        | 850   |
| 1     | V1   | 2000.00 | GLAZED VENTILATOR | 900    | 2000        | 850   |

# DOORS AND WINDOWS LEGENDS AND SCHEDULE

## 6)CAPITAL Height







Front View

















